Unit 1:computer network introduction

Computer network is a collection of interconnected devices that communicate with each other to exchange data and resources. It enables efficient communication and supports services like email, file sharing, and internet access.

Working

- A network consists of nodes such as computers, servers, routers, and switches that send or receive data.
- These nodes are connected through links, which may be wired (cables, optical fiber) or wireless (Wi-Fi, radio signals).
- When data is sent, it is broken into small packets and transmitted across the network.
- Protocols define how data packets are formatted, transmitted, received, and acknowledged.
- Each device is identified by a unique IP address, which ensures data reaches the correct destination.
- Network devices like switches and routers forward data packets along the best available path.
- Security mechanisms, such as firewalls, monitor traffic and allow or block data based on security rules.

Types of Computer Network Architecture

Computer Network falls under these broad Categories:

- **Client-Server Architecture:** Represents a type of computer network architecture in which nodes function as servers or

clients, where the server manages client behavior, known as Client-Server Architecture.

- **Peer-to-Peer Architecture:** Operates without any central server, allowing each device to act as either a client or a server, known as P2P (Peer-to-Peer) Architecture.

Network Devices/Network Hardware

Gate – Route – Smart – Dumb – Boost – Divide

(Gateway – Router – Switch – Hub – Repeater – Bridge)

Goals and Uses

- **Resource Sharing:** Allow multiple users to share hardware, software, and data resources efficiently.
- **Data Access and Sharing:** Provide easy access to stored data and allow information exchange between users and systems.
- **Reliability and Availability:** Improve system reliability using backup paths and fault-tolerant mechanisms.
- **Scalability and Growth:** Support easy expansion by adding new devices and services as demand increases.

Applications and Uses of Computer Network:

- Business applications (online banking, e-commerce)
- Education (online learning, sharing notes)
- Communication (email, chat, video calls)
- Entertainment (games, movies, music)
- Government services (online services)

Network Software

👉 **Network software is a set of programs and protocols that control, manage, and enable communication between devices in a computer network.**

Types of Connections

Point-to-Point:

- Uses a dedicated link to connect two devices
- Simple and easy to set up
- Limited to two devices only
- Does not require a network interface card (NIC) or a hub/switch
- Can become complex and difficult to manage as the network grows

Multipoint:

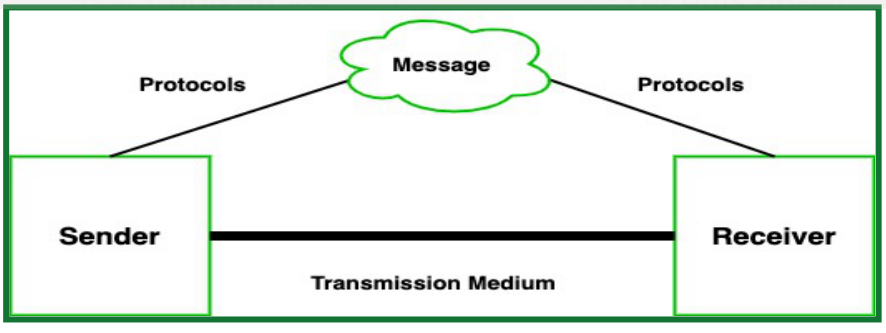
- Uses a single link to connect three or more devices
- More complex than point-to-point configuration
- Can be more efficient and cost-effective for larger networks
- Devices share the same link, which can lead to collisions and lower performance
- Commonly used in LANs and MANs

Data Communication - Definition, Components,Types, Channels Transferring data over a transmission medium between two or more devices, systems, or places is known as data communication.

Components of Data Communication

A communication system is made up of the following components:

1. **Message:** A message is a piece of information that is to be transmitted from one person to another. It could be a text file, an audio file, a video file, etc.
2. **Sender:** It is simply a device that sends data. It can be a computer, mobile, telephone, laptop, video camera, or workstation, etc.
3. **Receiver:** It is a device that receives messages. It can be a computer, telephone mobile, workstation, etc.
4. **Transmission Medium / Communication Channels:** Communication channels are the medium that connect two or more workstations. Workstations can be connected by either wired media or wireless media.
5. **Protocol:** When someone sends the data (The sender), it should be understandable to the receiver also otherwise it is meaningless. For example, Sonali sends a message to Chetan. If Sonali writes in Hindi and Chetan cannot understand Hindi, it is a meaningless conversation.



Protocol type

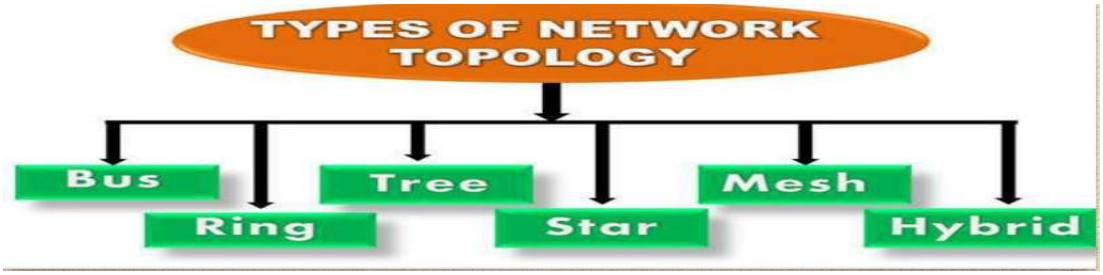
1 TCP(Transmission Control Protocol): It is responsible for dividing messages into packets on the source computer

2 IP(Internet Protocol) IP is responsible for handling the address of the destination computer so that each packet is sent to its proper destination.

Type of data communication

As we know that data communication is communication in which we can send or receive data from one device to another. The data communication is divided into three types:

1. Simplex Communication: It is one-way unidirectional communication in which one device only receives and another device only sends data and devices uses their entire capacity in transmission. For example, IoT, entering data using a keyboard, listening music using a speaker, etc.
2. Half Duplex communication: It is a two-way bidirectional communication in which both the devices can send and receive data but not at the same time. When one device is sending data then another device is only receiving and vice-versa. example, walkie-talkie.
3. Full-duplex communication: It is a two-way bidirectional communication in which both the devices can send and receive data at the same time. example, mobile phones, landlines, etc.



Types of Network Topology

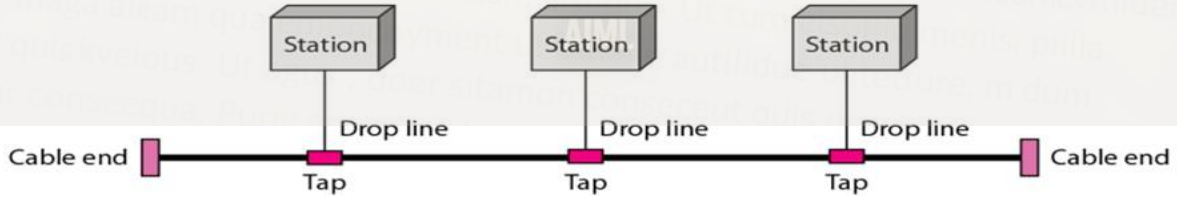
Network topology is the arrangement of devices in a computer network. It shows how computers, servers, and other devices are connected and how data flows between them. Choosing the right topology is important because it affects the performance, cost, reliability, and security of the network. There are two main types of topology:

- Physical Topology: The actual physical layout of cables and devices.
- Logical Topology: How data moves across the network, regardless of physical layout.

All the following topologies (Bus, Star, Mesh, Ring, Tree, Hybrid) are physical topologies. Logical topology refers to how data flows in the network, such as CSMA/CD in Ethernet or token passing in ring networks.

Bus Topology

Bus Topology is a network type in which every computer and network device is connected to a single cable. It is bi-directional. It is a multi-point connection and a non-robust topology because if the backbone fails the topology crashes.



Advantages

- Simple and easy to install
- Low cost (less cable used)

- Suitable for small networks

🔥 Disadvantages

- If main cable fails → whole network stops
- More traffic → slow speed
- Difficult to find faults

Applications

1️⃣ Small LAN Networks

- Used in small offices or computer labs

2️⃣ Temporary Networks

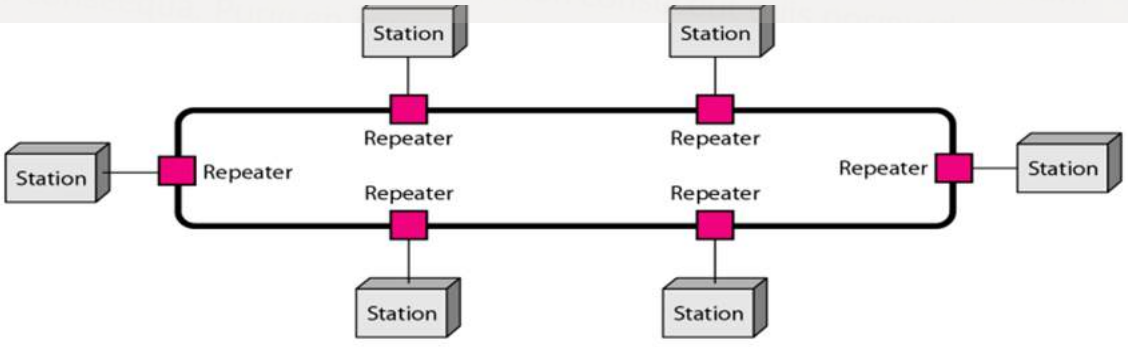
- Used for short-term setups (like events, labs)

3️⃣ Home Networks (Old systems)

- Earlier used in small home connections

Ring Topology

👉 **Ring topology is a network in which all devices are connected in a circular (ring) form, where each device is connected to two other devices.**



🔥 Working

- Devices are connected in a circle
- Data travels in **one direction**
- Each device receives data and passes it to the next
- The process continues until it reaches the destination

👉 Simple:
👉 **"Circle → data moves one by one → reaches destination"**

🔥 Advantages

- No data collision
- Equal access for all devices
- Better performance than bus in some cases

🔥 Disadvantages

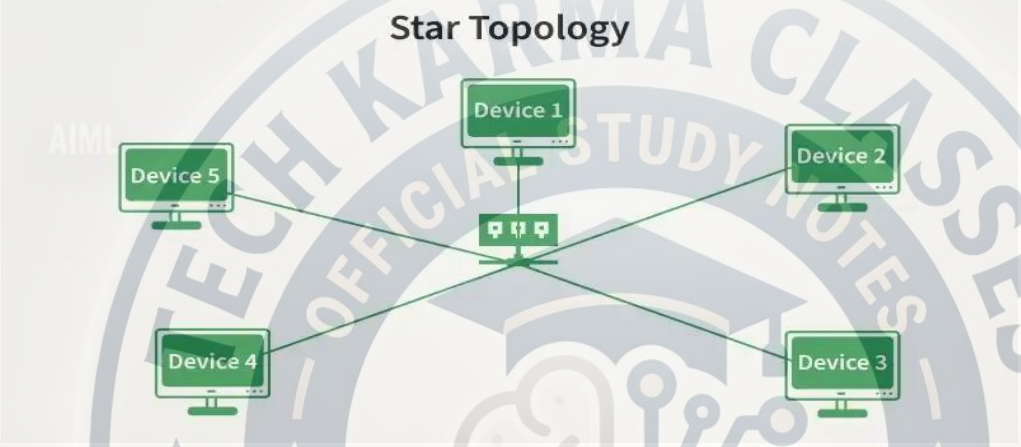
- If one device fails → whole network affected
- Difficult to troubleshoot
- Adding/removing devices is hard

🔥 Applications

- Used in older LAN systems
- Token Ring networks
- Some fiber networks

Star Topology

👉 **Star topology is a network in which all devices are connected to a central device like a hub or switch.**



🔥 Working

- All devices connect to a **central hub/switch**
- When a device sends data → it goes to the central device
- Central device forwards data to the correct destination

👉 Simple:
👉 **"Device → Center → Correct Device"**

🔥 Advantages

- Easy to install and manage
- Failure of one device does not affect others
- Easy to detect faults

🔥 Disadvantages

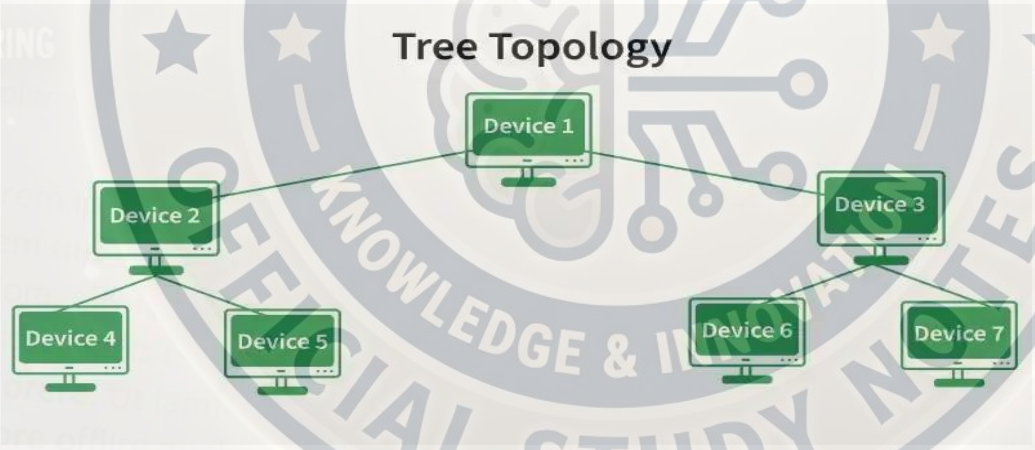
- If central device fails → whole network stops
- Cost is higher (more cable required)

🔥 Applications

- Modern LAN networks
- Office networks
- Home WiFi networks

Tree Topology

👉 **Tree topology is a network structure in which multiple star networks are connected to a main backbone in a hierarchical (tree-like) form.**



🔥 Working

- It combines **Star + Bus topology**
- There is a **main backbone cable**
- Multiple star networks are connected to it
- Data flows from top (root) to lower levels (nodes)

👉 Simple:
👉 **"Tree structure → root → branches → devices"** 🌳

🔥 Advantages

- Better management due to hierarchy
- Easy to expand (add new nodes)
- Suitable for large networks

🔥 Disadvantages

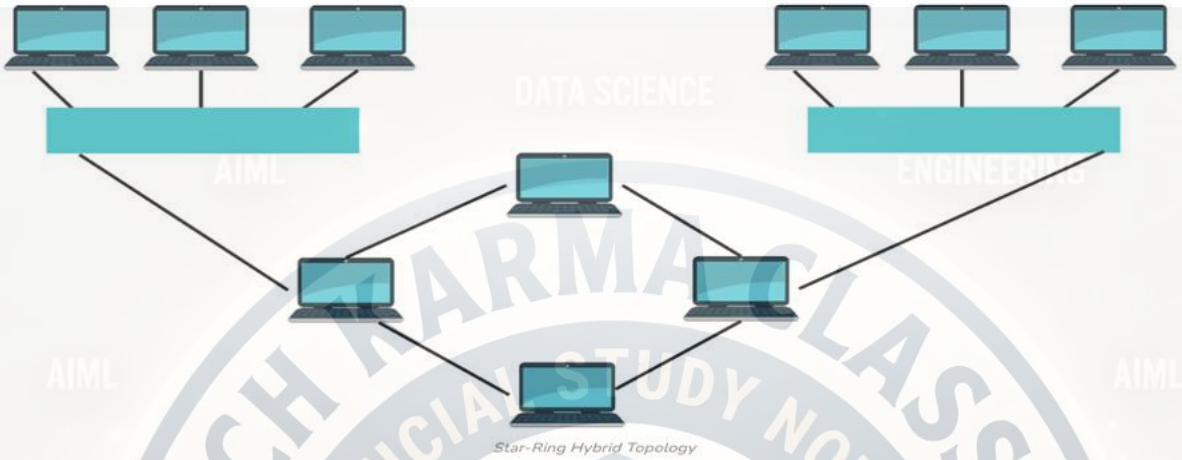
- Backbone failure → large part of network affected
- Complex setup
- Cost higher than bus

🔥 Applications

- Large organizations
- Universities
- Corporate networks

Hybrid Topology (Definition)

👉 **Hybrid topology is a network that combines two or more different types of topologies (like star, bus, ring, etc.) into one network.**



🔥 Working

- Different topologies are connected together
- Each part works according to its own topology
- All parts are connected to form a single network

👉 Simple:
👉 **"Mix of topologies → one network"**

🔥 Advantages

- Suitable for large networks
- Very flexible (can design as needed)
- Reliable (failure in one part doesn't affect whole network)

🔥 Disadvantages

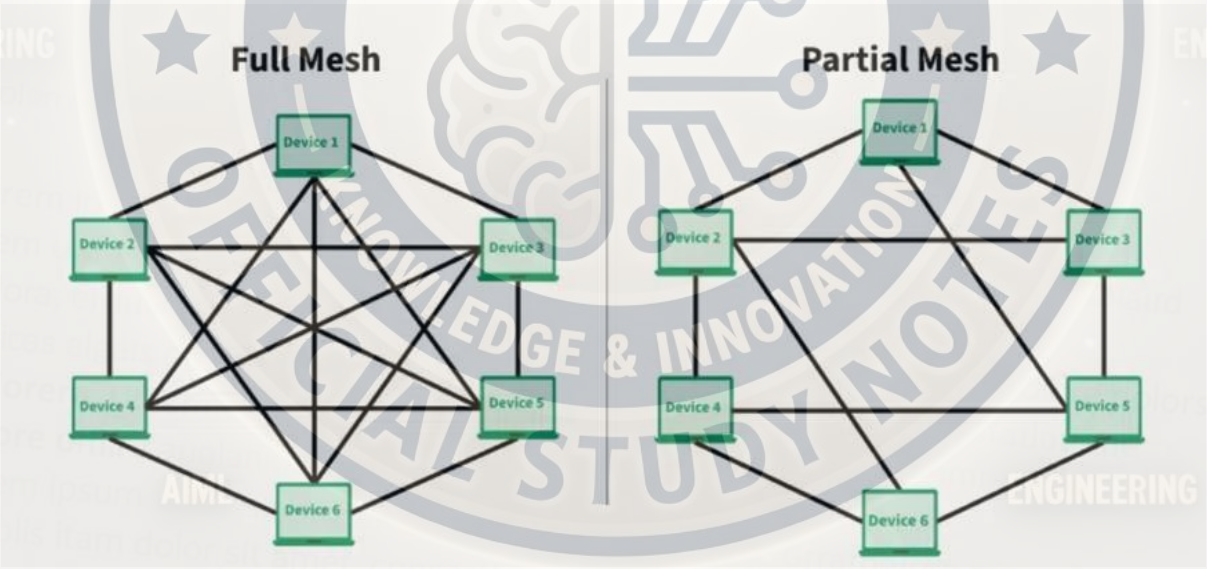
- Complex design
- High cost
- Difficult to manage

🔥 Applications

- Large companies
- Banks
- Universities
- Data centers

Mesh Topology (Definition)

👉 **Mesh topology is a network in which every device is directly connected to every other device.**



🔥 Working

- Each device has a **dedicated link** with every other device
- Data travels through a **direct path** (no need of central device)
- Multiple paths are available for communication

👉 Simple:
👉 **“Everyone connected to everyone”**

🔥 Types of Mesh

1️⃣ **Full Mesh**

👉 Every device connected to all others

2️⃣ **Partial Mesh**

👉 Only some devices are fully connected

🔥 Advantages

- Highly reliable (if one link fails, others work)
- No data loss
- Fast communication

🔥 Disadvantages

- Very expensive (lots of cables)
- Complex installation
- Difficult to manage

🔥 Applications

- Military networks
- Banking systems
- Critical systems (high security required)

Types of Computer Networks

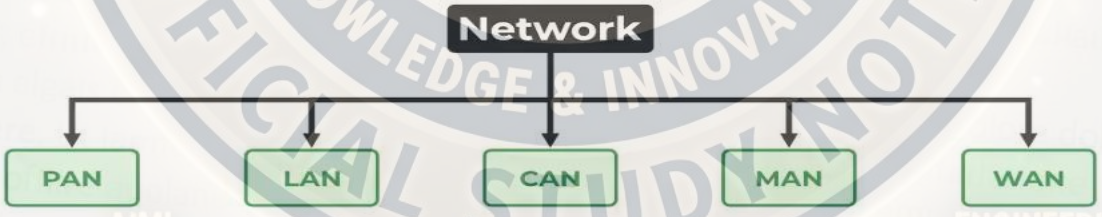
A computer network is a group of two or more independent computers and devices connected to share data, resources, and services. These

connections can be established using wired media or wireless technologies. Types of Computer Networks are classified based on several factors:

1. Geographical area
2. Transmission medium
3. Ownership and access control

Classification Based on Geographical Area

Categorizing computer networks according to the physical distance or area they cover, ranging from small personal spaces to large global regions. Based on geographical area, computer networks are mainly divided into five types:

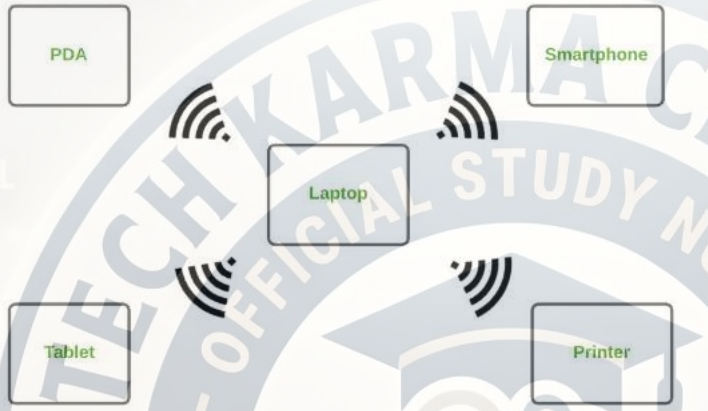


1. Personal Area Network (PAN)

This connects personal devices within a very short range around a single user.

- **Coverage Range:** 1–10 meters
- **Key Technologies:** Bluetooth, NFC, Infrared
- **Real-World Example:** Smartphone connected to wireless earbuds

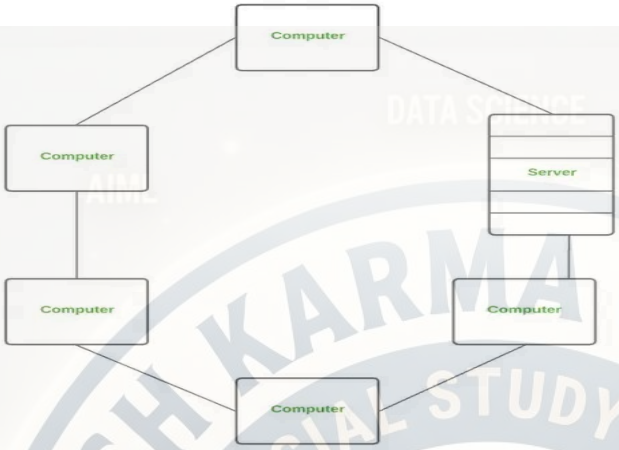
- **When It Is Used:** For short-range communication between personal devices



2. Local Area Network (LAN)

This connects computers and devices within a limited area such as a home, office, or building.

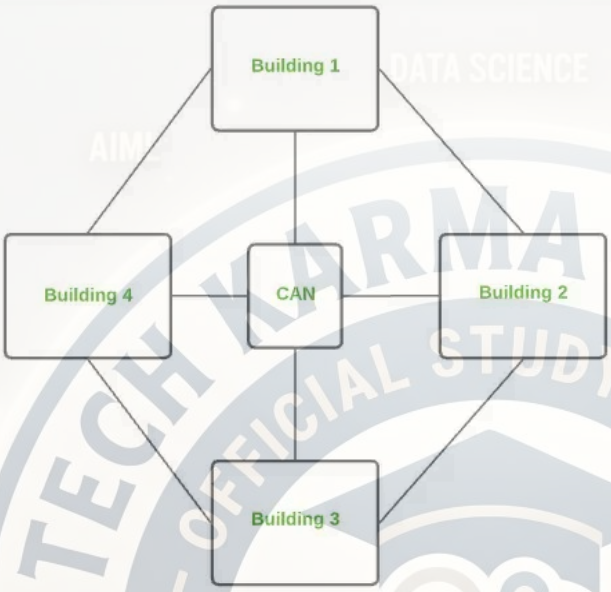
- **Coverage Range:** One room, building, or small campus(Approx range: **100 meter se kuch kilometer tak**)
- **Key Technologies:** Ethernet, Wi-Fi
- **Real-World Example:** Home or office Wi-Fi network
- **When It Is Used:** For fast and secure communication within a local location



3. Campus Area Network (CAN)

A CAN connects multiple LANs within a campus or a group of nearby buildings.

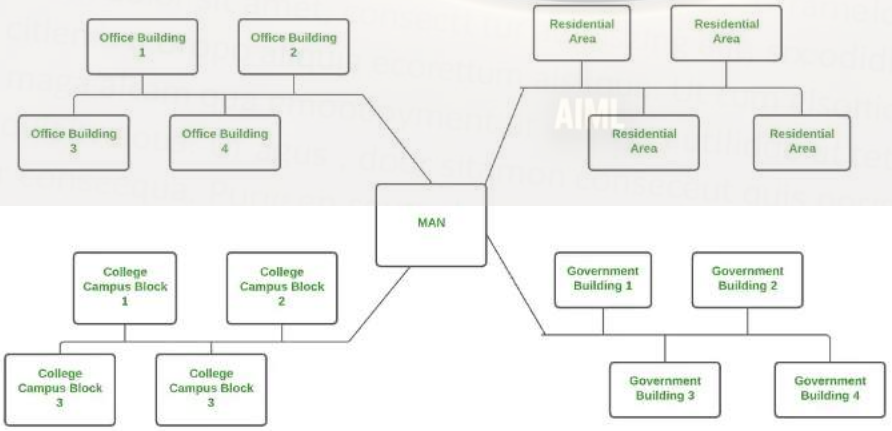
- **Coverage Range: 1–5 km** University or corporate campus
- **Key Technologies:** Ethernet, Fiber Optics
- **Real-World Example:** College or corporate campus network
- **When It Is Used:** To interconnect LANs across a campus under one organization



4. Metropolitan Area Network (MAN)

A MAN connects multiple networks within a city or metropolitan region.

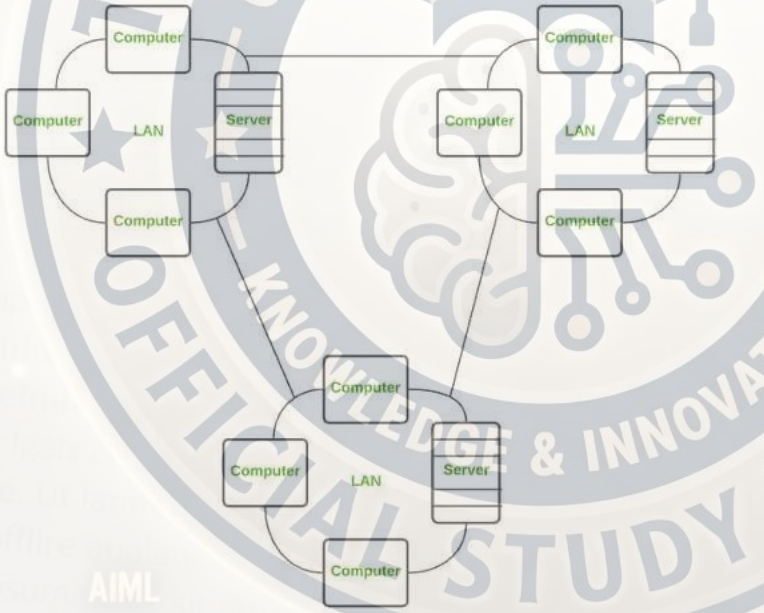
- ***Coverage Range:*** *5–50 km City or large town*
- ***Key Technologies:*** *Fiber Optics, Microwave, Metro Ethernet*
- ***Real-World Example:*** *City-wide ISP network*
- ***When It Is Used:*** *To provide high-speed connectivity across a city*



5. Wide Area Network (WAN) Wide Area Network

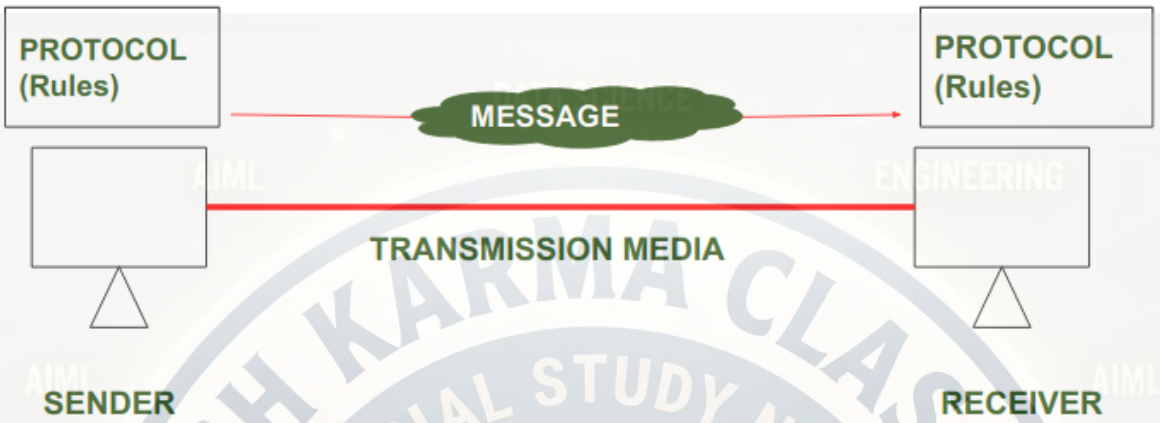
A WAN connects networks over large geographical areas such as countries or continents.

- ***Coverage Range:*** *Country, continent, or global*
- ***Key Technologies:*** *Leased lines, Satellite links, Internet*
- ***Real-World Example:*** *The Internet*
- ***When It Is Used:*** *To enable long-distance and global communication*



What is Protocol?

A protocol is a set of rules that determines how data is sent and received over a network. The protocol is just like a language that computers use to talk to each other, ensuring they understand and can respond to each other's messages correctly.



Key Elements of Protocol

- ***Syntax:*** *Syntax refers to the structure or the format of the data. The starting 8 bits of data are considered as the address of the sender. The next 8 bits are considered to be the address of the receiver. The remaining bits are considered as the message itself.*
- ***Semantics:*** *Semantics defines data transmitted between devices. It provides rules and norms for understanding message or data element values and actions.*
- ***Timing:*** *Timing refers to the synchronization and coordination between devices while transferring the data. Timing ensures at what time data should be sent and how fast data can be sent. For example, If a sender sends 100 Mbps but the receiver can only handle 1 Mbps, the receiver will overflow and lose data.*
- *Timing ensures preventing of data loss, collisions, and other timing-related issues.*
- ***Sequence Control:*** *Sequence control ensures the proper ordering of data packets. Through this mechanism, the data is delivered in correct order.*

Types of Protocol

1)Network Layer Protocols : *Network layer protocol is known as the Layer 3 of the network architecture.*

- *It is responsible for packet routing, forwarding, and addressing of data packets throughout the network.*
- *IP and ICMP are the network layer protocols.*

2)Transport Layer Protocols*: it provides end-to-end service ensuring data transfer across apps on different devices.*

- *TCP and UDP are the most popular transport layer protocols.*

3) Application Layer Protocol: *it provides communication between applications running on different devices.*

- *HTTP, FTP, and SMTP are examples.*

4)Wireless Protocols: *it is basically used in wireless communication which enables data transfer through wireless networks. Bluetooth, Wi-Fi,*

5)Routing Protocols: *Routing protocol establishes the best/optimal network pathways throughout the network for fastest data transmission. Routers share information to develop and maintain routing tables.*

RIP, OSPF, and BGP are examples.

6)Security Protocols : *it protects data confidentiality, integrity, and authenticity while transmission of data over the network. They include SSL and TLS, encryption methods, and authentication protocols for providing data security.*

7)Internet Protocols : *IP identifies devices uniquely.*

Internet protocol provides data communication through routing and forwarding data packets from one device to another by unique addressing scheme.

Important Protocols Used in Computer Network

Here are some key protocols that are widely used in computer networks:

- ***TCP (Transmission Control Protocol):*** *Ensures data is sent and received accurately by breaking it into packets, sending them, and reassembling them at the destination.*
- ***IP (Internet Protocol):*** *Addresses and routes the packets to make sure they reach the right destination.*
- ***HTTP/HTTPS (HyperText Transfer Protocol/Secure):*** *HTTP used for transferring web pages on the internet. When you browse a website, your browser uses HTTP to request and display web pages. And HTTPs is a secure version of HTTP that encrypts data to protect it from being intercepted.*
- ***FTP (File Transfer Protocol):*** *Used for transferring files between computers on a network. It allows users to upload and download files.*
- ***SMTP (Simple Mail Transfer Protocol):*** *Used for sending emails. It transfers emails from a client to a server or between servers.*
- ***DNS (Domain Name System):*** *It is used to translates human-readable domain names (like www.example.com) into IP*

addresses that computers use to identify each other on the network.

- ***DHCP (Dynamic Host Configuration Protocol):*** *Automatically assigns IP addresses to devices on a network, ensuring each device has a unique address.*
- ***SSH (Secure Shell):*** *Provides a secure way to access and manage devices over a network. It encrypts the data, making it safe from eavesdropping.*
- ***SNMP (Simple Network Management Protocol):*** *Used for managing and monitoring network devices like routers, switches, and servers. It collects and organizes information about these devices*

COMPUTER NETWORK PROTOCOLS

A PROTOCOL is a set of rules and standards that controls and enables communication between devices in a network.

TYPES OF PROTOCOLS

1	NETWORK LAYER PROTOCOLS	Works at Layer 3 of the OSI model. Responsible for logical addressing, routing, forwarding and delivery of data packets across networks.	EXAMPLES: IP, ICMP • IP – Internet Protocol • ICMP – Internet Control Message Protocol
2	TRANSPORT LAYER PROTOCOLS	Works at Layer 4. Provides end-to-end communication between applications on different devices. Ensures data delivery, reliability and error control.	EXAMPLES: TCP, UDP • TCP – Transmission Control Protocol (connection-oriented, reliable) • UDP – User Datagram Protocol (connectionless, faster)
3	APPLICATION LAYER PROTOCOLS	Works at Layer 7. Provides communication between applications running on different devices and enables user services.	EXAMPLES: HTTP, FTP, SMTP • HTTP – HyperText Transfer Protocol • FTP – File Transfer Protocol • SMTP – Simple Mail Transfer Protocol
4	WIRELESS PROTOCOLS	Used in wireless communication. Enables data transfer through wireless networks over short or long range.	EXAMPLES: Bluetooth, Wi-Fi • Bluetooth – For short range communication • Wi-Fi (IEEE 802.11) – For wireless LAN communication
5	ROUTING PROTOCOLS	Establishes the best/optimal paths in a network for fastest and most efficient data transmission. Routers exchange information and maintain routing tables.	EXAMPLES: RIP, OSPF, BGP • RIP – Routing Information Protocol • OSPF – Open Shortest Path First • UDP – Border Gateway Protocol
6	SECURITY PROTOCOLS	Protects data confidentiality, integrity and authenticity during transmission over the network. Includes encryption, authentication and secure communication.	EXAMPLES: SSL/TLS, Encryption, Authentication • SSL/TLS – Secure Sockets Layer / Transport Layer Security • Encryption Methods – AES, RSA etc. • Authentication Protocols – Kerberos
7	INTERNET PROTOCOLS	Defines rules for internetworking. IP identifies devices uniquely and provides data communication through routing and forwarding using unique addressing.	EXAMPLES: IP • IP – Internet Protocol (IPv4, IPv6)

IMPORTANT PROTOCOLS USED IN COMPUTER NETWORKS

TCP (Transmission Control Protocol)
Ensures data is sent and received accurately. Breaks data into packets, sends them, and reassembles them at the destination.

IP (Internet Protocol)
Addresses and routes the packets to make sure they reach the right destination.

HTTP / HTTPS
HTTP is used for transferring web pages on the internet. HTTPS is the secure version of HTTP that encrypts data to protect it from interception.

FTP (File Transfer Protocol)
Used for transferring files between computers on a network. It allows users to upload and download files.

SMTP (Simple Mail Transfer Protocol)
Used for sending emails. It transfers emails from a client to a server or between servers.

DNS (Domain Name System)
Translates human-readable domain names (like www.example.com) into IP addresses that computers use to identify each other.

DHCP (Dynamic Host Configuration Protocol)
Automatically assigns IP addresses to devices on a network, ensuring each device has a unique address.

SSH (Secure Shell)
Provides a secure way to access and manage devices over a network. It encrypts the data, making it safe from eavesdropping.

SNMP (Simple Network Management Protocol)
Used for managing and monitoring network devices like routers, switches, and servers. It collects and organizes information about these devices.

KEY POINTS TO REMEMBER

- ✔ Protocols define HOW data is transmitted, received and understood.
- ✔ They ensure reliable, secure and efficient communication.
- ✔ Different layers use different protocols to perform their specific tasks.

MEMORY TIP

Net - Trans - App - Wire - Route - Secure - Internet
(1 to 7 Types Order)

HOW DATA COMMUNICATION WORKS (IN SHORT)



Protocols work at different layers to make this communication possible.

PROTOCOLS & OSI MODEL (QUICK VIEW)

7. Application Layer	Application Layer Protocols (HTTP, FTP, SMTP, DNS...)
4. Transport Layer	Transport Layer Protocols (TCP, UDP)
3. Network Layer	Network Layer Protocols (IP, ICMP)
2. Data Link Layer	(MAC, LLC – Framing, Physical Addressing)
1. Physical Layer	(Bits Transmission over Physical Medium)

What is Standards?

Standards are the set of rules for data communication that are needed for the exchange of information among devices. It is important to follow Standards which are created by various Standard Organizations like IEEE, ISO, ANSI, etc.

Types of Standards

- ***De Facto Standard:*** *The meaning of the word " De Facto " is " By Fact " or "By Convention". These are the standards that have not been approved by any Organization but have been adopted as Standards because of their widespread use. Also, sometimes these standards are often established by Manufacturers.*

  For example : *Apple and Google are two companies that established their own rules for their products which are different. Also, they use some same standard rules for manufacturing their products.*

- ***De Jure Standard:*** *The meaning of the word "De Jure" is "By Law" or "By Regulations".Thus, these are the standards that have been approved by officially recognized bodies like ANSI, ISO, IEEE, etc. These are the standards that are important to follow if it is required or needed.*

  For example : *All the data communication standard protocols like SMTP, TCP, IP, UDP etc. are important to follow the same when we need them.*

Layered Architecture

a model where a whole network process is divided into smaller sub-tasks. then assigned to a specific layer to perform only the dedicated tasks.

Elements of Layered Architecture

- **Service:** it is defined as a set of functions and tasks being provided by a lower layer to a higher layer. Each layer performs a different type of task. Therefore, actions provided by each layer are different.
- **Protocol:** Protocol is defined as a set rule used by the layer for exchanging and transmission of data with its peer entities.

  Interface: Interface is defined as a channel that allows to transmit the messages from one layer to the another.

Evolution of Layered Architecture

In computer networks, layered architecture is majorly used for communication. The two network models that makes use of layered architecture are:

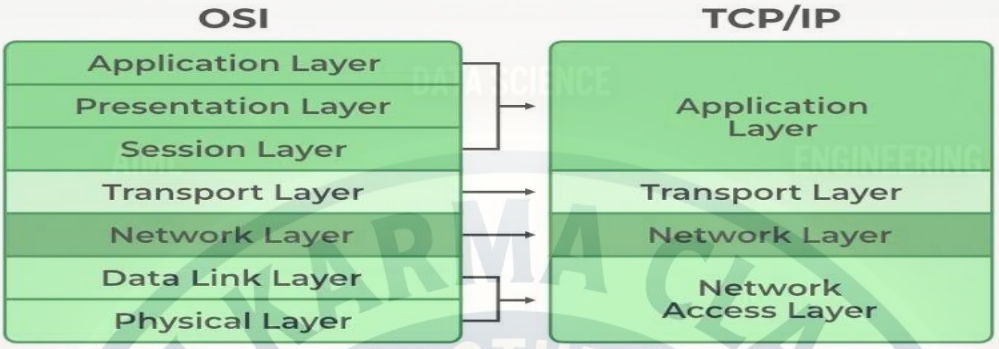
- OSI Model
- TCP/IP Model

1. OSI Model

OSI stands for Open Systems Interconnection. OSI is a seven layered architecture. All these seven layers work collaboratively to transmit data from one layer to another.

- **Physical Layer:** Physical layer is the lowest layer of OSI model and is responsible for the physical connection between all the required devices. The information present in physical layer is in the form of bits. Physical layer performs various functions such as bit rate control, bit synchronization, transmission mode etc.

- **Data Link Layer:** Data Link layer provides with successful delivery of message from one node to the another. It checks whether this delivery of message is error free. Other functions performed by data link layer are error control, framing, flow control etc.
- **Network Layer:** Network Layer is responsible for the transmission of data from one host to another host. It performs other tasks such routing and logical addressing.
- **Transport Layer:** Transport Layer is defined as a layer that takes services from network layer and provides services to application layer. Other tasks performed by transport layer are service point addressing, segmentation and reassembling.
- **Session Layer:** Session layer is defined as a layer that is responsible for establishing a connection, maintenance of session and to provide with security. Other functions of session Layer are to establish session, termination and synchronization.
- **Presentation Layer:** The data from application layer is extracted at the presentation layer. This layer is also known as translation layer. The functions of presentation layer are encryption, decryption, compression and translation.
- **Application Layer:** Application layer is the topmost layer of OSI Model. Application layer is also known as desktop layer. It provides with other functions such as directory services, mail services, network virtual terminal etc.



2. TCP/IP Model

1. Network Access Layer

It is a group of applications requiring network communications. This layer is responsible for generating the data and requesting connections. It acts on behalf of the sender and the Network Access layer on the behalf of the receiver. During this article, we will be talking on the behalf of the receiver.

2. Internet Layer

This layer parallels the functions of OSI's Network layer. It defines the protocols which are responsible for the logical transmission of data over the entire network.

- **IP:** IP stands for Internet Protocol and it is responsible for delivering packets from the source host to the destination host IP has 2 versions: IPv4 and IPv6. IPv4 is the one that most websites are using currently. But IPv6 is growing as the number of IPv4 addresses is limited in number when compared to the number of users.
- **ICMP:** ICMP stands for Internet Control Message Protocol. It is encapsulated within IP datagrams and is responsible for providing hosts with information about network problems.

- **ARP:** ARP stands for Address Resolution Protocol. Its job is to find the hardware address of a host from a known IP address.

3. Transport Layer

The TCP/IP transport layer protocols exchange data receipt acknowledgments and retransmit missing packets to ensure that packets arrive in order and without error. Transmission Control Protocol (TCP) and User Datagram Protocol are transport layer protocols at this level (UDP).

- **TCP:**. TCP transmits data in a way that resembles character-by-character transmission rather than separate packets. A starting point that establishes the connection, the whole transmission in byte order, and an ending point that closes the connection make up this transmission.
- **UDP:** The datagram delivery service is provided by UDP, the other transport layer protocol. Connections between receiving and sending hosts are not verified by UDP. Applications that transport little amounts of data use UDP rather than TCP because it eliminates the processes of establishing and validating connections.

4. Application Layer

This layer is analogous to the transport layer of the OSI model. It is responsible for end-to-end communication and error-free delivery of data. It shields the upper-layer applications from the complexities of data. The three main protocols present in this layer are:

- **HTTP and HTTPS:** HTTP stands for Hypertext transfer protocol. It is used by the World Wide Web to manage communications between web browsers and servers. HTTPS stands for HTTP-Secure. It is a combination of HTTP with SSL(Secure

Socket Layer). It is efficient in cases where the browser needs to fill out forms, sign in, authenticate, and carry out bank transactions.

- **SSH:** SSH stands for Secure Shell. It is a terminal emulations software similar to Telnet. The reason SSH is preferred is because of its ability to maintain the encrypted connection. It sets up a secure session over a TCP/IP connection.
- **NTP:** NTP stands for Network Time Protocol. It is used to synchronize the clocks on our computer to one standard time source. It is very useful in situations like bank transactions. Assume the following situation without the presence of NTP. Suppose you carry out a transaction, where your computer reads the time at 2:30 PM while the server records it at 2:28 PM. The server can crash very badly if it's out of sync.

Benefits of Layered Architecture

- Modularity
- Interoperability
- Flexibility
- Reusability
- Scalability
- Security

Challenges in Layered Architecture

- Performance Overhead
- Complexity in Implementation
- Resource Utilization
- Debugging and Troubleshooting

- Protocol Overhead

Network Devices/Connecting devices

Hub vs. Switch vs. Router

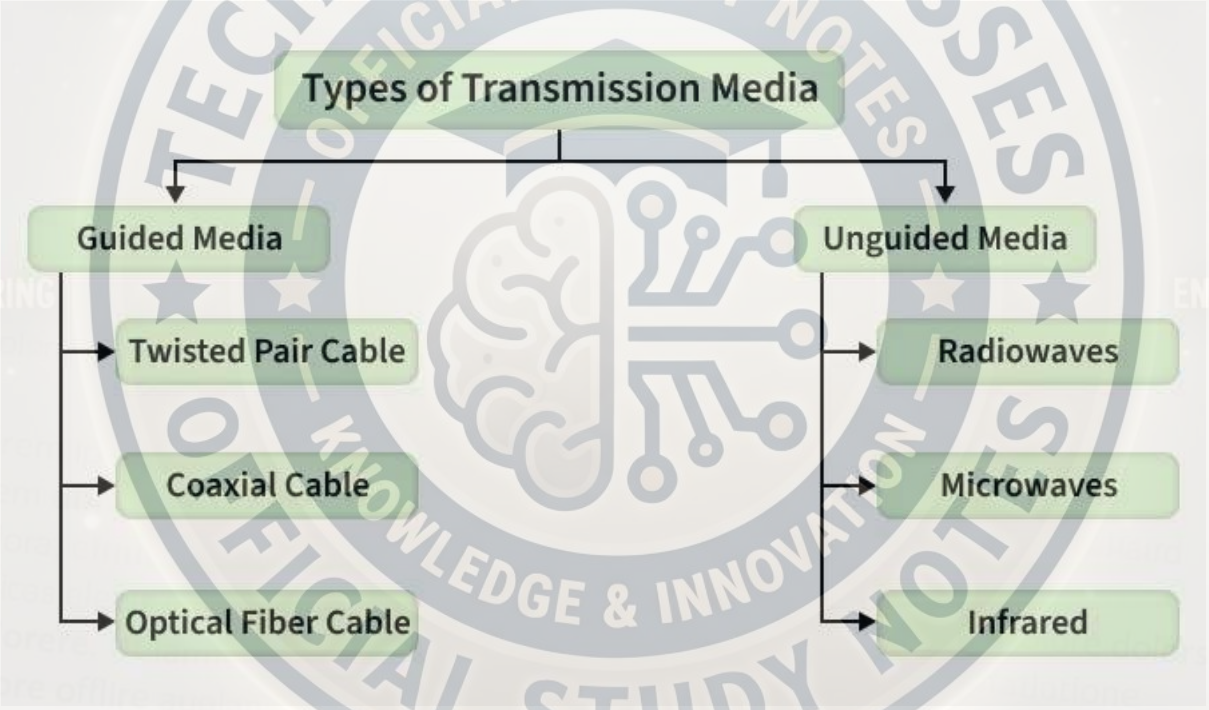
Feature	Hub	Switch	Router
Layer	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 3 (Network)
Address Type	None (Bits)	MAC Address	IP Address
Data Flow	Broadcast (to all)	Unicast (to destination)	Route (best path)
Intelligence	Dumb	Smart	Smarter
Used For	Obsolete	Connecting Devices (LAN)	Connecting Networks (WAN)

Transmission Media in Computer Networks

Transmission media refers to the physical or wireless communication channel used to carry data signals from one device to another

The selection of a transmission medium depends on factors such as:

- Transmission distance
- Data transfer speed (bandwidth)
- noise
- Cost and installation requirements



1. Twisted Pair Cable

It is consists of two individually insulated copper conductors twisted together in a helical pattern to minimize electromagnetic interference (EMI) and crosstalk between adjacent pairs. It is most commonly used and cost-effective

- Twisting of conductors helps reduce interference and signal degradation.
- Multiple pairs can be bundled together within a single cable.

- Widely used due to low cost, flexibility, and ease of installation.
- Commonly used in telephone systems and local area networks (LANs).

(a) Unshielded Twisted Pair (UTP): it is consists of twisted copper wire pairs without any additional metallic shielding. The twisting provides basic protection against interference, making UTP suitable for environments with moderate noise levels.



Advantages:

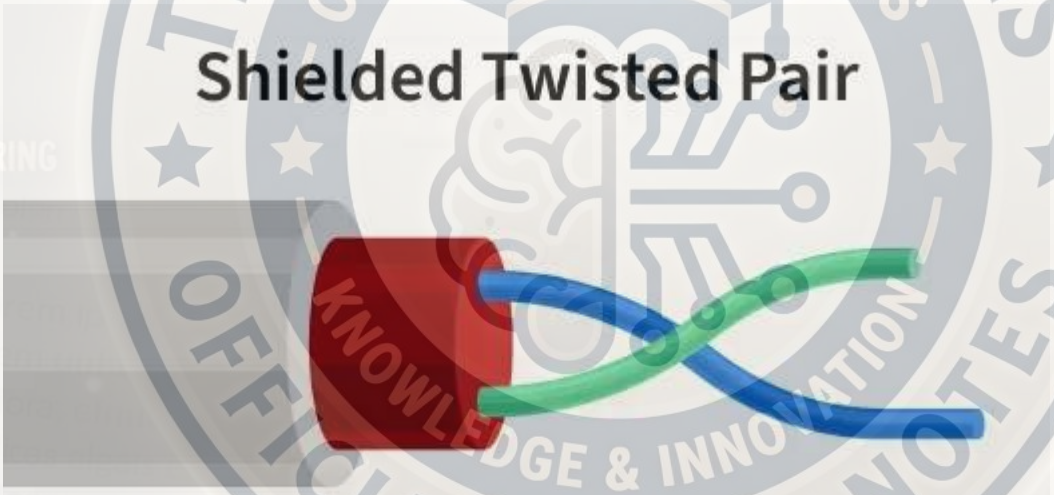
- Least expensive among guided transmission media.
- Easy to install and maintain due to its lightweight and flexible nature.
- Supports high data transmission rates over short distances.
- Widely used in Ethernet-based LANs and telephone networks.

Disadvantages:

- Lower performance and capacity compared to Shielded Twisted Pair (STP).

- Limited transmission distance due to signal attenuation.
- More susceptible to noise and electromagnetic interference.

(b) Shielded Twisted Pair (STP): Shielded Twisted Pair (STP) cable consists of twisted pairs of insulated copper conductors enclosed within an additional metallic shielding layer, such as a foil or braided shield. This shielding provides enhanced protection against external electromagnetic interference (EMI) and reduces crosstalk, resulting in more stable and reliable data transmission.



Advantages

- Provides better performance at higher data transmission rates compared to UTP.
- Significantly reduces electromagnetic interference and crosstalk.
- Ensures more reliable and stable communication, especially in noisy environments.
- Suitable for high-speed Ethernet networks and voice and data transmission systems.

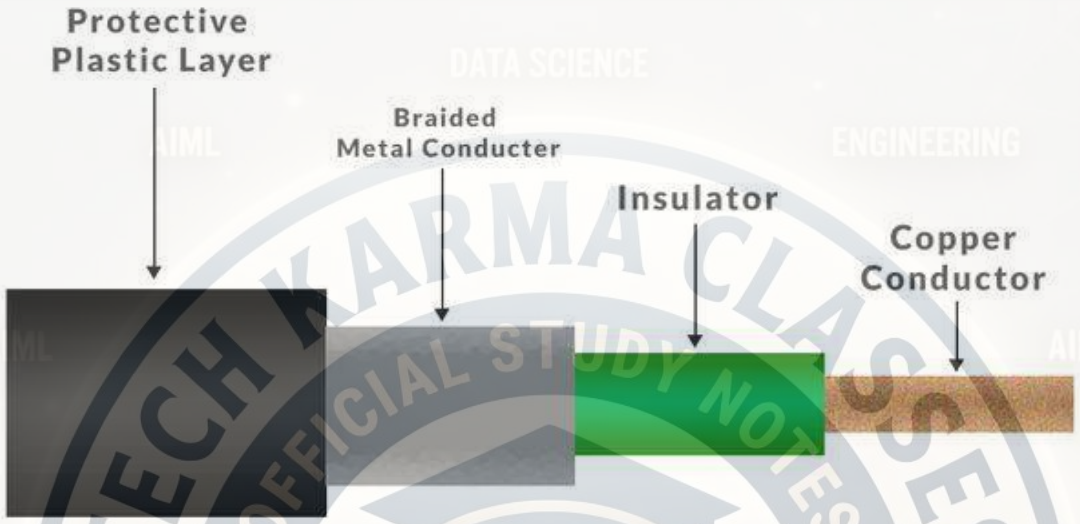
Disadvantages

- Higher cost compared to Unshielded Twisted Pair (UTP).
- More complex installation and termination due to shielding requirements.
- Bulkier and less flexible, making cable management more difficult.

2. Coaxial Cable

Coaxial Cable consists of a central copper conductor surrounded by a dielectric insulating layer, a metallic shielding layer, and an outer protective jacket. This structure allows signals to be transmitted with better protection against noise and interference compared to twisted pair cables.

- Supports data transmission in both baseband mode (single channel) and broadband mode (multiple channels).
- Provides higher bandwidth than twisted pair cables.
- Strong shielding makes it more resistant to noise, crosstalk, and electromagnetic interference (EMI).
- Commonly used in cable television (CATV), broadband networks, and analog television systems.
- More durable and reliable due to its layered construction.
- Easier to install and maintain compared to optical fiber cables.



Advantages:

- Supports higher bandwidth than twisted pair cables.
- Offers better signal quality and reliability.
- Allows multiple channels using frequency division multiplexing (FDM).
- Less affected by external electromagnetic interference.

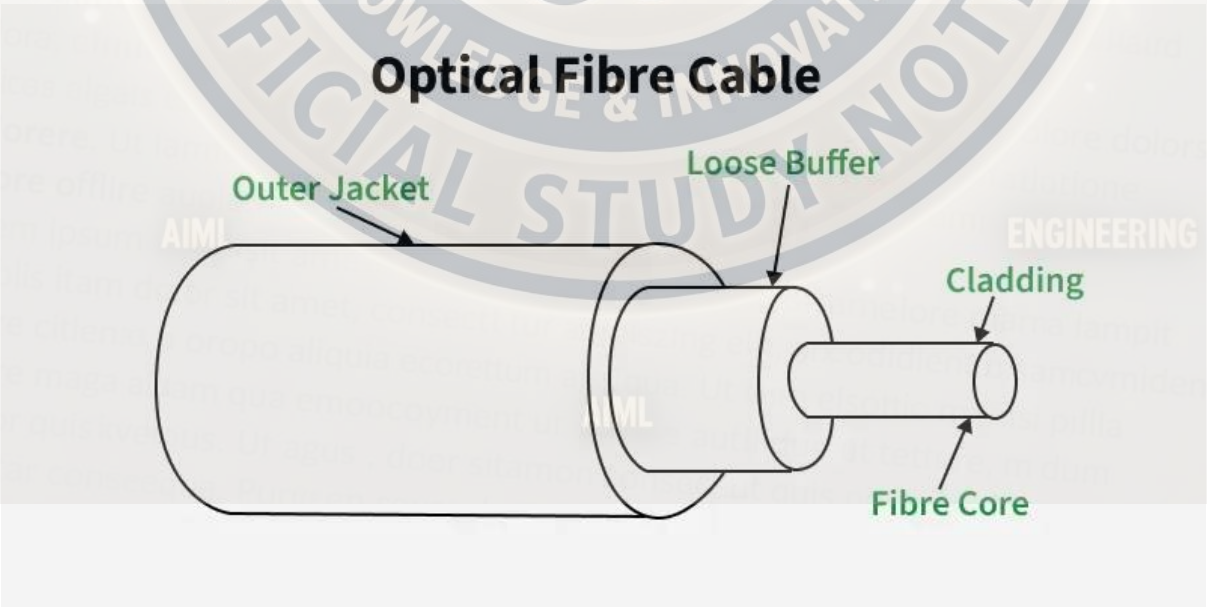
Disadvantages:

- More expensive than twisted pair cables.
- Requires proper grounding to ensure safety and minimize interference.
- Bulkier and less flexible due to multiple layers.
- More vulnerable to security breaches, as the cable can be physically tapped.

3. Optical Fiber Cable

It is a guided transmission medium that transmits data in the form of light signals through a glass or plastic core using the principle of total internal reflection. The core is surrounded by a cladding layer with a lower refractive index, which confines the light within the core and enables high-speed, long-distance data transmission.

- Supports very large data volumes at extremely high speeds.
- Can operate in unidirectional or bidirectional communication modes.
- WDM (Wavelength Division Multiplexer) allows multiple light signals to be transmitted simultaneously over a single fiber.
- Widely used where high bandwidth, long distance, and low signal loss are required.
-



Advantages:

- Provides very high bandwidth and data-carrying capacity.
- Lightweight and compact, making installation easier over long distances.

- Exhibits low signal attenuation, suitable for long-distance communication.
- Immune to electromagnetic interference (EMI) and electrical noise.
- Resistant to corrosion and harsh environmental conditions.

Disadvantages:

- Complex installation and maintenance due to precise splicing and handling requirements.
- High initial installation and equipment cost.
- Fragile compared to copper cables and requires careful handling.

Applications:

- **Medical:** Used in endoscopy, imaging systems, and laser-based surgical procedures.
- **Defense and Aerospace:** Supports secure, high-speed, and interference-free communication.
- **Communication:** Widely used in internet backbone networks, undersea cables, and telecommunication systems.
- **Industrial and Automotive:** Used for sensing, lighting, data communication, and safety systems.

Unguided Media

It is also known as wireless or Unbounded transmission media , uses electromagnetic waves to transmit data without any physical medium. Signals propagate through free space such as air or vacuum.

Features:

- Signals propagate through air or free space
- Less secure due to broadcast nature
- Suitable for long-distance communication

1. Radio Waves

Radio waves are a type of electromagnetic wave that can easily be generated and can propagate through buildings and other obstacles. They do not require line-of-sight between transmitting and receiving antennas, making them highly suitable for broadcast communication and wireless data transmission.

Frequency Range:
3 kHz – 300 GHz

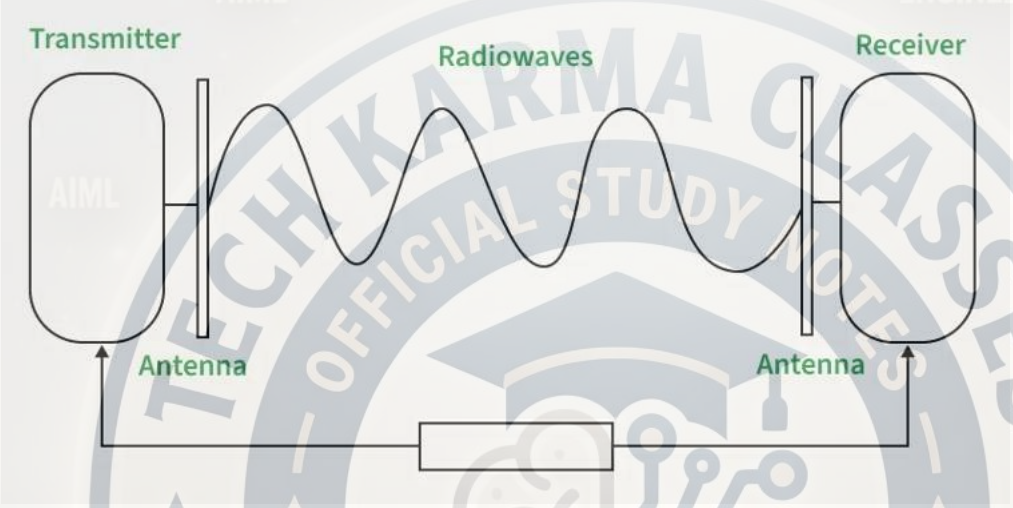
Applications:
AM and FM radio broadcasting, television transmission, cordless phones, and wireless communication.

Types of Radio Waves:

- **Shortwave:** AM radio broadcasting
- **VHF (Very High Frequency):** FM radio and television
- **UHF (Ultra High Frequency):** Television and mobile communication

Components:

- **Transmitter:** Generates and modulates the signal
- **Receiver:** Receives and demodulates the signal



2.

Microwaves

Micro waves are a form of unguided transmission media that use line-of-sight communication, where the transmitting and receiving antennas must be properly aligned. The transmission range depends on the height of the antennas. Microwaves operate in the frequency range of 1 GHz to 300 GHz and are widely used in mobile communication, satellite links, and television distribution.

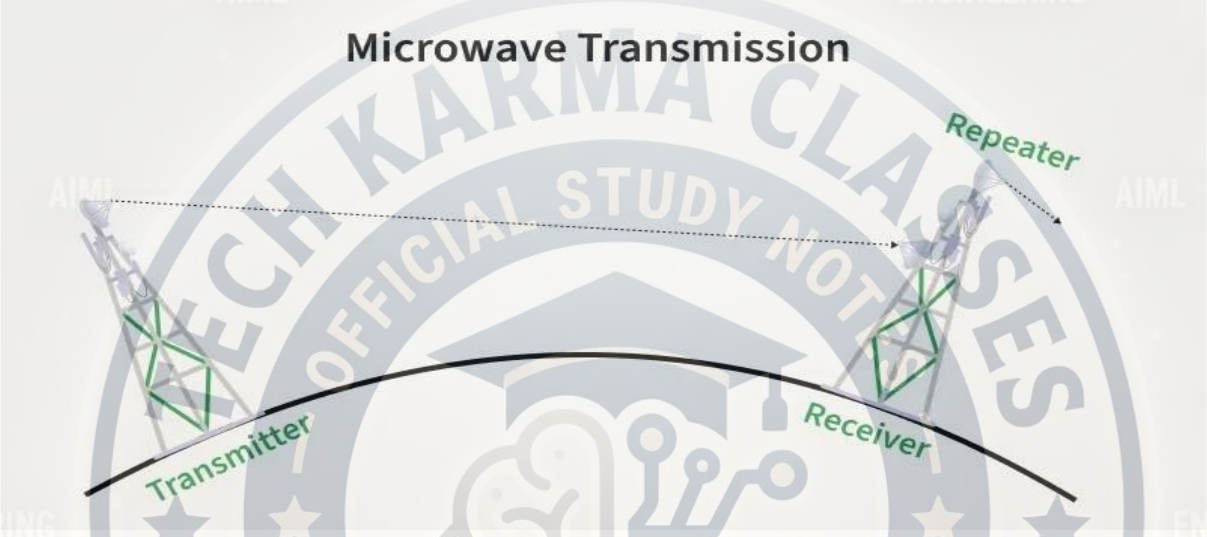
Advantages:

- Cost-effective compared to laying physical cables
- No need for land acquisition
- Suitable for communication over difficult terrains and oceans
- Supports high data transmission rates

Disadvantages:

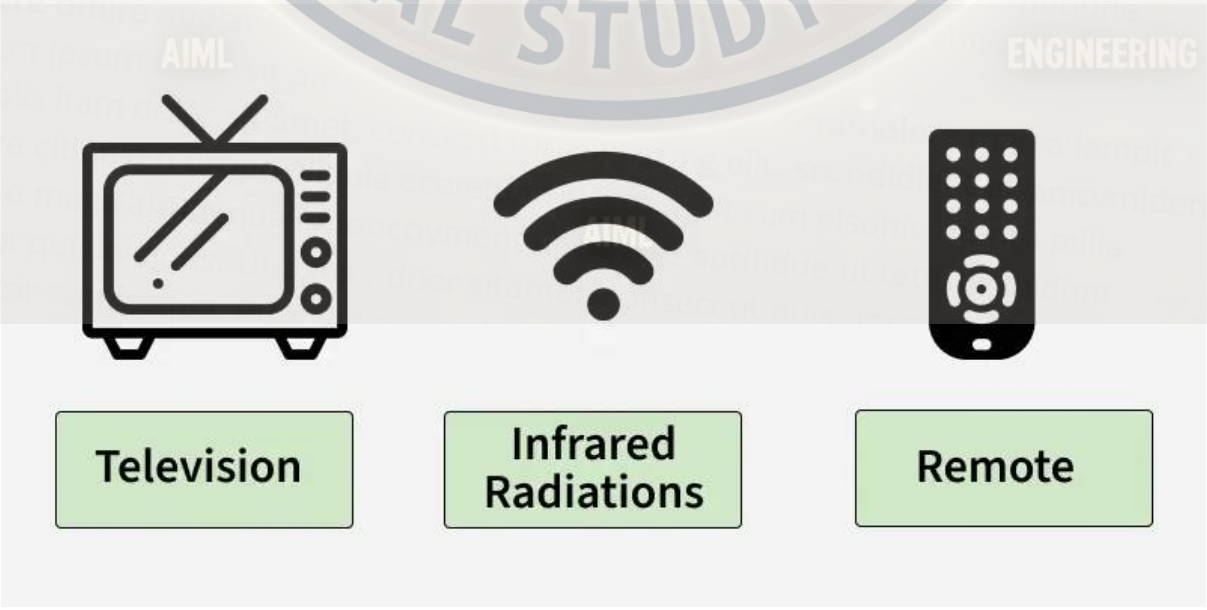
- Less secure without proper encryption
- Susceptible to weather conditions such as rain and fog

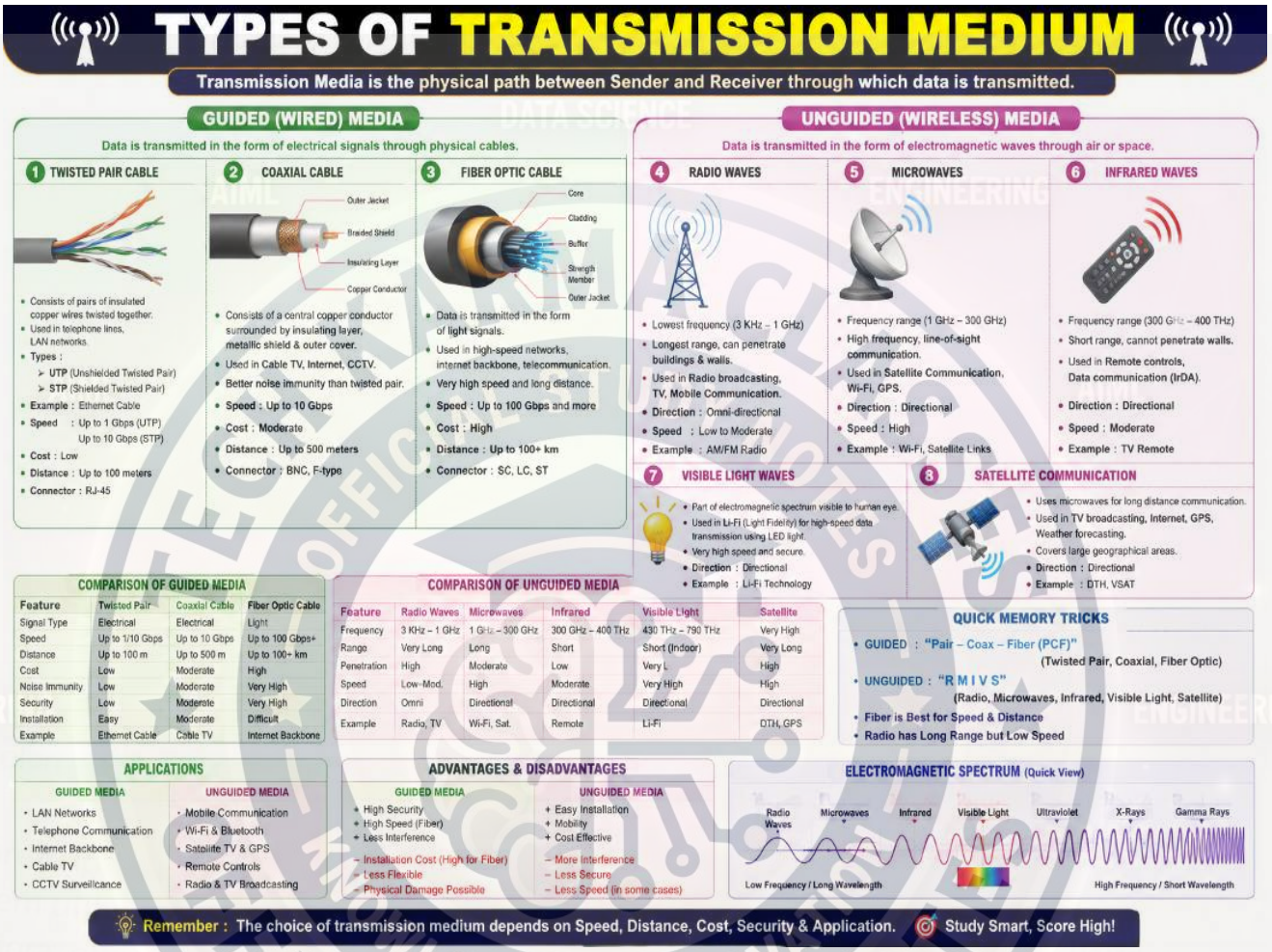
- Affected by obstacles due to line-of-sight requirement
- High cost of antenna design, installation, and maintenance



3. Infrared

Infrared waves are used for short-range wireless communication and operate in the frequency range of 300 GHz to 400 THz. They cannot penetrate solid obstacles, which limits their range but helps minimize interference between nearby systems. Infrared waves are commonly used in TV remote controls, wireless keyboards, mice, and printers.



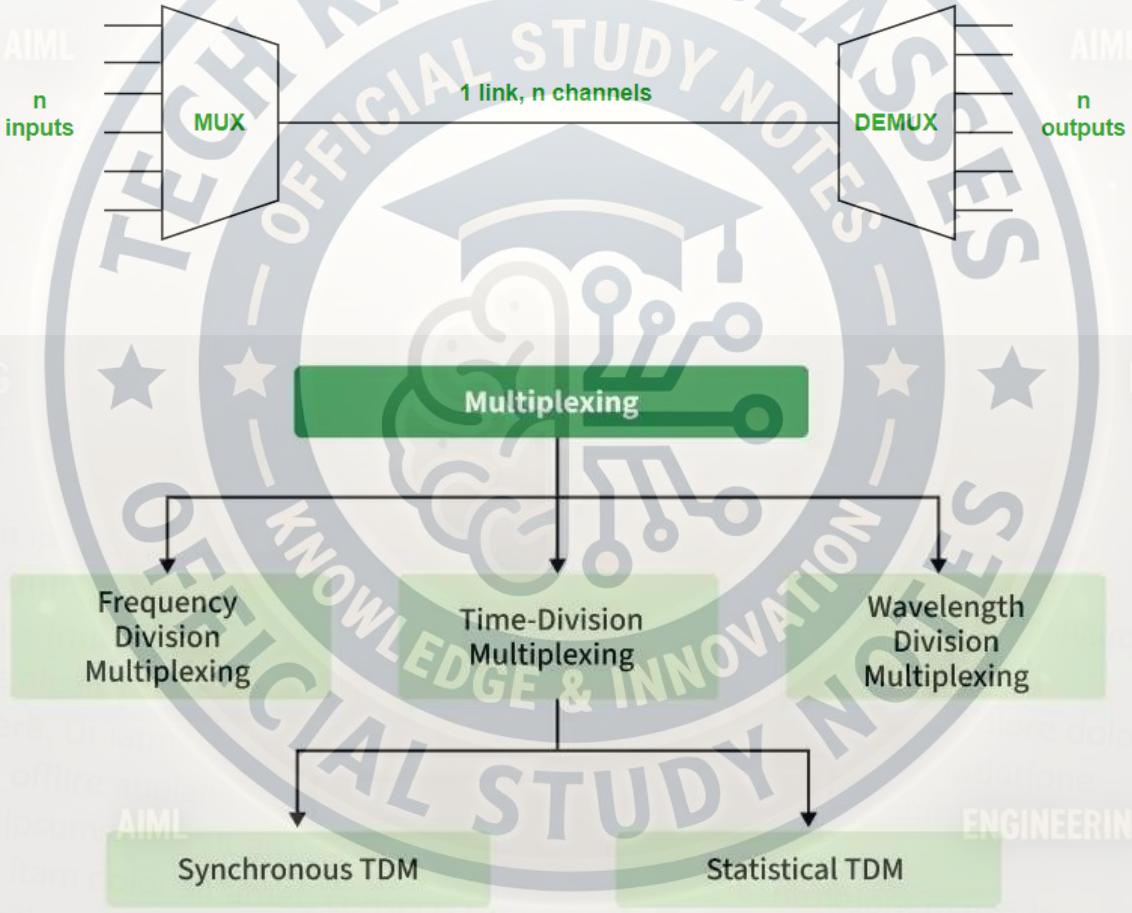


Difference Between Radio Waves, Micro Waves, and Infrared Waves

Feature	Radio Waves	Microwaves	Infrared Waves
Frequency	Lowest	Medium	Higher
Range	Very Long	Long	Short
Speed of Data	Low to Moderate	High	Moderate
Direction	Omni-directional	Directional	Directional
Can Pass Through Walls	Yes	Limited	No
Communication Type	Broadcasting	Point-to-point	Short distance
Used In	Radio, TV, Mobile	Wi-Fi, Satellite, GPS	TV Remote, Sensors
Frequency Range	3 KHz – 1 GHz	1 GHz – 300 GHz	300 GHz – 400 THz
Example	FM Radio	Wi-Fi	Remote Control
Cost	Low	Moderate	Low

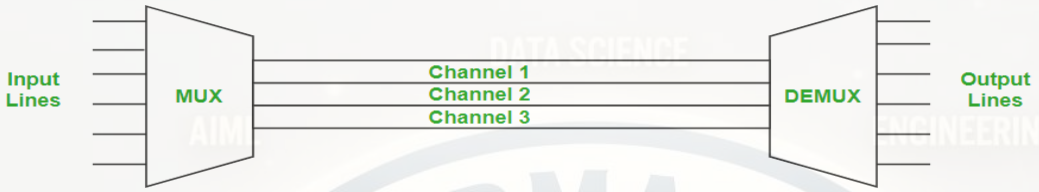
Types of Multiplexing in Data Communications

Multiplexing in data communications is a method that combines multiple signals or data streams into one signal over a shared medium. This process allows for efficient use of resources and can significantly increase the amount of data that can be sent over a network

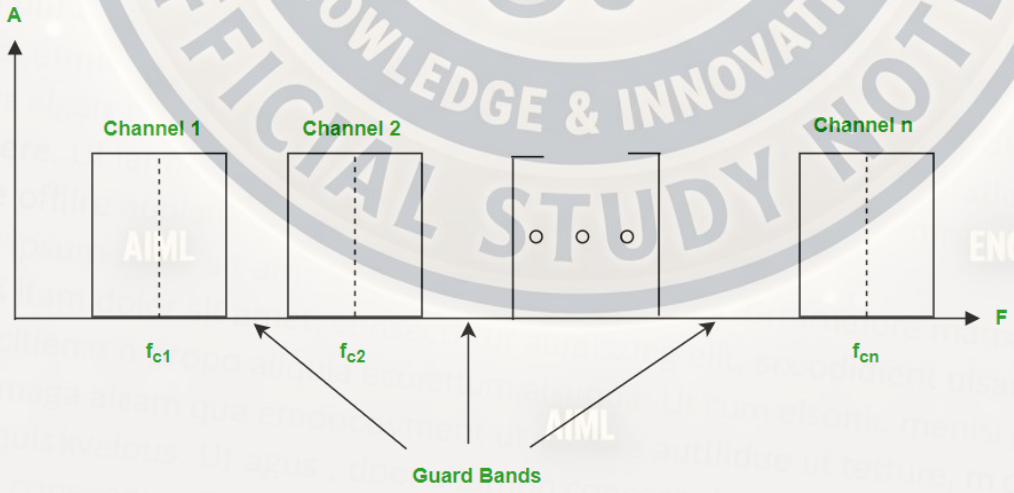


1. Frequency Division Multiplexing

it is defined as a type of multiplexing where the bandwidth of a single physical medium is divided into a number of smaller, independent frequency channels.



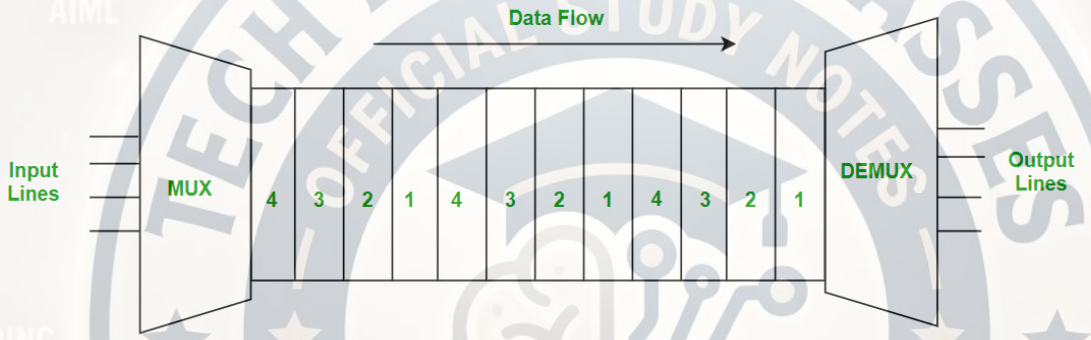
- In FDM, we can observe a lot of inter-channel cross-talk because in this type of multiplexing the bandwidth is divided into frequency channels.
- In order to prevent the inter-channel cross talk, unused strips of bandwidth must be placed between each channel. These unused strips between each channel are known as guard bands.



Note: *Frequency Division Multiplexing is used in radio and television transmission.*

2. Time Division Multiplexing

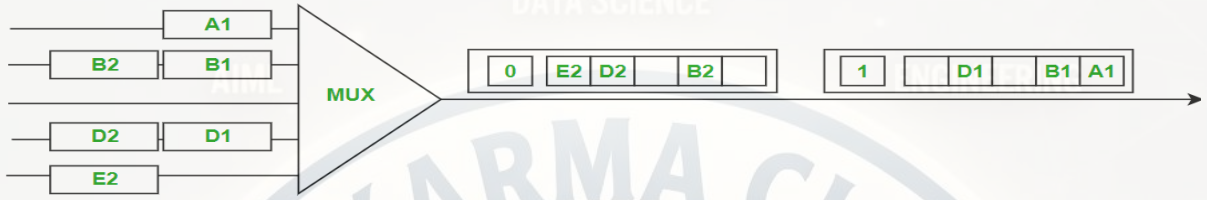
Time-division multiplexing is multiplexing wherein FDM, instead of sharing a portion of the bandwidth in the form of channels, in TDM, time is shared. Each connection occupies a portion of time in the link. In Time Division Multiplexing, all signals operate with the same frequency (bandwidth) at different times.



Note: *There are two Types of Time Division Multiplexing - Synchronous Time Division Multiplexing & Statistical (or Asynchronous) Time Division Multiplexing.*

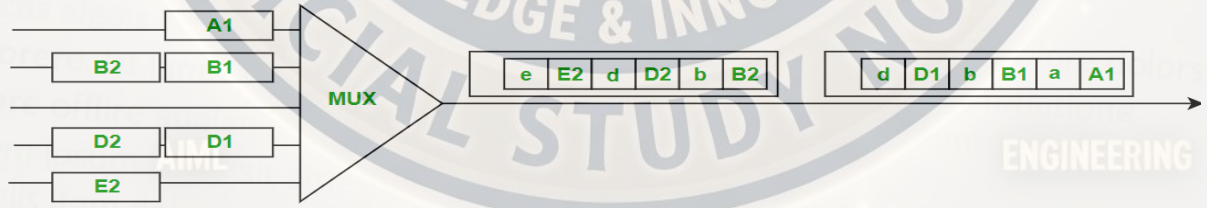
2.1 Synchronous TDM : Synchronous TDM is a type of Time Division Multiplexing where the input frame already has a slot in the output frame.

- Time slots are grouped into frames. One frame consists of one cycle of time slots.
- Synchronous TDM is not efficient because if the input frame has no data to send, a slot remains empty in the output frame.
- In this, we need to mention the synchronous bit at the beginning of each frame.



2.2 Statistical TDM: Statistical TDM is a type of Time Division Multiplexing where the output frame collects data from the input frame till it is full not leaving an empty slot like in Synchronous TDM.

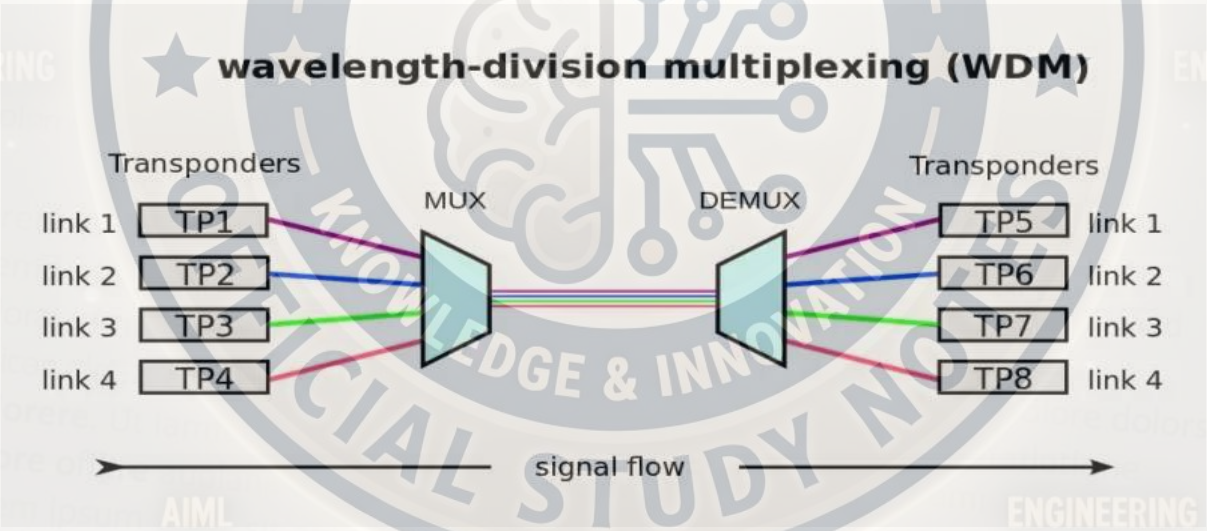
- In this, we need to include the address of each particular data in the slot that is being sent to the output frame.
- Statistical TDM is a more efficient type of time-division multiplexing as the channel capacity is fully utilized and improves the bandwidth efficiency.



3. Wavelength Division Multiplexing

Wavelength Division Multiplexing (WDM) is a multiplexing technology used to increase the capacity of optical fiber by transmitting multiple optical signals simultaneously over a single optical fiber, each with a different wavelength.

- Each signal is carried on a different wavelength of light, and the resulting signals are combined onto a single optical fiber for transmission.
- At the receiving end, the signals are separated by their wavelengths, demultiplexed and routed to their respective destinations.
- It is used in telecommunications, cable TV, ISPs, and data centers for high-speed, long-distance data transmission.



WDM

WDM can be divided into two categories:

- Dense Wavelength Division Multiplexing (DWDM) is used to multiplex a large number of optical signals onto a single fiber, typically up to 80 channels with a spacing of 0.8 nm or less between the channels.
- Coarse Wavelength Division Multiplexing (CWDM) is used for lower-capacity applications, typically up to 18 channels with a spacing of 20 nm between the channels.

Read about Differences between DWDM and CWDM.

Advantages over Time Division Multiplexing (TDM):

- Higher data rates & capacity
- Lower power consumption
- Reduced equipment complexity
- Flexible & easily upgradable

Advantages of Multiplexing

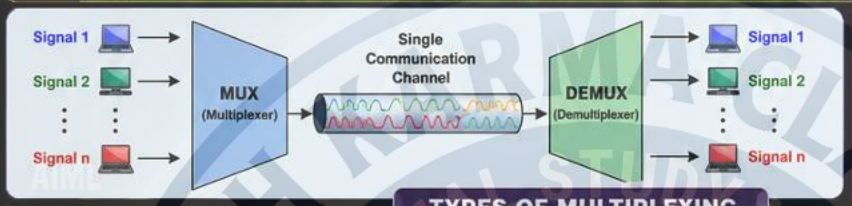
- **Efficient Use of Bandwidth:** You can send more than one signal over a single channel. This way, you can use the channel's capacity more efficiently.
- **Increased Data Transmission**: Multiplexing can significantly boost the amount of data that can be sent over a network simultaneously, enhancing overall transmission capacity.
- **Scalability**: Multiplexing allows networks to easily expand and accommodate more data streams without requiring significant changes to the existing infrastructure.
- **Flexibility**: Different types of multiplexing (TDM, FDM, WDM, CDM) can be used based on the specific needs and characteristics of the communication system, providing flexibility in network design.

Disadvantages of Multiplexing

- **Synchronization Issues**: Ensuring that multiple data streams remain properly synchronized can be challenging, leading to potential data loss or errors if not managed correctly.
- **Latency**: Combining multiple signals into one can introduce delays, as each data stream needs to be processed, synchronized, and demultiplexed at the receiving end.
- **Signal Degradation**: Over long distances, multiplexed signals can experience degradation and interference, requiring additional measures such as signal boosters or repeaters to maintain quality.
- **Resource Management**: Allocating and managing resources for multiplexing can be complicated, requiring careful planning and real-time adjustments to avoid congestion and ensure efficient operation.

TYPES OF MULTIPLEXING IN DATA COMMUNICATIONS

Multiplexing is a technique of combining multiple signals from different sources and transmitting them together over a single communication channel.
At the receiver side, **Demultiplexing** is used to separate the individual signals.



WHY MULTIPLEXING?

- ✓ Improves efficiency of the channel
- ✓ Reduces cost
- ✓ Saves bandwidth
- ✓ Allows simultaneous transmission
- ✓ Better utilization of communication media

TYPES OF MULTIPLEXING

1. FDM – FREQUENCY DIVISION MULTIPLEXING

In FDM, the entire available bandwidth of the channel is divided into several sub-channels of different frequency bands. Each signal is transmitted simultaneously over its own frequency band.

How it Works

- The available bandwidth is divided into frequency bands.
- Each signal is modulated on a different carrier frequency.
- All signals are transmitted simultaneously.
- At the receiver, bandpass filters separate the signals.



Example



Advantages

- ✓ All signals can be sent at the same time.
- ✓ Simple for analog signals.
- ✓ High-speed transmission.

Disadvantages

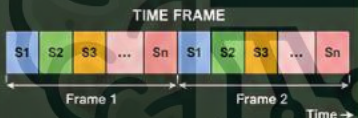
- ✗ Guard bands are required between channels which wastes some bandwidth.
- ✗ Expensive filters are required.
- ✗ Not suitable for digital signals.

2. TDM – TIME DIVISION MULTIPLEXING

In TDM, the entire time of the channel is divided into repetitive time slots. Each signal is transmitted during its assigned time slot. Only one signal is transmitted at a time.

How it Works

- Time is divided into frames.
- Each frame is divided into n time slots (where n = number of signals).
- Each signal gets its own time slot.
- Signals are sent in a cyclic order.
- At the receiver, signals are separated based on their time slots.



Example



Advantages

- ✓ No guard band required.
- ✓ Better for digital signals.
- ✓ Efficient use of bandwidth.

Disadvantages

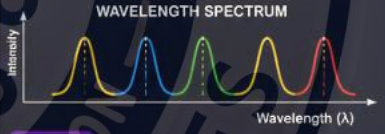
- ✗ More complex than FDM.
- ✗ Requires precise timing and synchronization.
- ✗ If one time slot is not used, bandwidth is wasted.

3. WDM – WAVELENGTH DIVISION MULTIPLEXING

In WDM, multiple signals are transmitted simultaneously over a single optical fiber using different wavelengths (colors) of light.

How it Works

- Different signals are modulated on different wavelengths.
- All wavelengths are combined and sent through the same fiber.
- At the receiver, a demultiplexer separates the wavelengths.



Example



Advantages

- ✓ Very high bandwidth.
- ✓ Low signal loss.
- ✓ High-speed transmission over long distance.

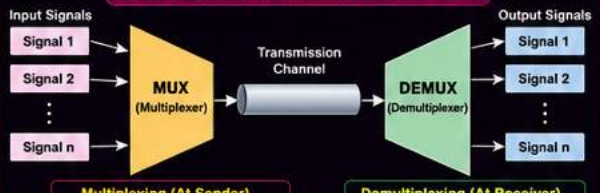
Disadvantages

- ✗ Expensive equipment.
- ✗ More complex design.
- ✗ Used only in optical fiber communication.

COMPARISON TABLE

Feature	FDM	TDM	WDM
Full Form	Frequency Division Multiplexing	Time Division Multiplexing	Wavelength Division Multiplexing
Divide By	Frequency Bands	Time Slots	Wavelengths (Light Colors)
Signals Sent	Simultaneously	One at a Time (in different slots)	Simultaneously
Used In	Radio, TV, Cable TV	Telephone Systems, GSM, ISDN	Fiber Optic Communication
Medium	Coaxial Cable, Radio Spectrum	Twisted Pair, Digital Links	Optical Fiber
Bandwidth Efficiency	Low (Due to Guard Bands)	High	Very High
Complexity	Low to Moderate	Moderate	High
Cost	Moderate	Low to Moderate	High
Best For	Analog Signals	Digital Signals	High-Speed Long Distance

MULTIPLEXING & DEMULTIPLEXING FLOW



Multiplexing (At Sender)
Combines multiple input signals into a single composite signal.

Demultiplexing (At Receiver)
Separates the composite signal into original individual signals.

KEY POINTS TO REMEMBER

- ★ Multiplexing increases the capacity of a communication channel.
- ★ It reduces the number of physical links required.
- ★ FDM divides bandwidth, TDM divides time, WDM divides wavelength.
- ★ Demultiplexing is the reverse process of multiplexing.
- ★ WDM provides the highest data rate and is used in modern high-speed networks.

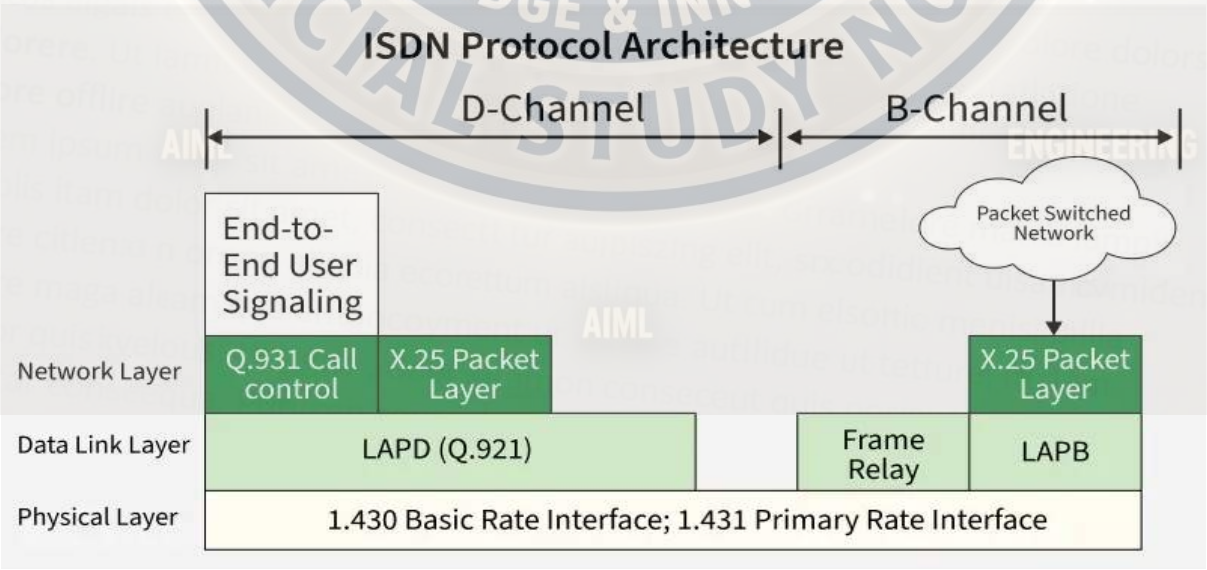
QUICK MEMORY TRICK

FDM = Frequency mein baanto
TDM = Time mein baanto
WDM = Wavelength (Light) mein baanto

Integrated Services Digital Network (ISDN)

The Integrated Services Digital Network (ISDN) is a set of communication protocols that enable the simultaneous digital transmission of voice, video, data, and other network services over traditional Public Switched Telephone Network(PSTN) connections. ISDN aims to enable end-to-end digital connectivity to serve a diverse set of services.

- It provides a packet-switched connection for data in increments of 64 kilobit/s.
- It provided a maximum of 128 kbit/s bandwidth in both upstream and downstream directions.
- A greater data rate was achieved through channel bonding.
- Generally, ISDN B-channels of three or four BRIs (six to eight 64 kbit/s channels) are bonded.



ISDN Protocol Architecture

Note: *ISDN is a circuit-switched telephone network system, but it also provides access to packet-switched networks that allow digital transmission of voice and data. This results in potentially better voice or data quality than an analog phone can provide.*

Working of ISDN

The ISDN works based on the standards defined by ITU-T (formerly CCITT). The Telecommunication Standardization Sector (ITU-T) coordinates standards for telecommunications on behalf of the International Telecommunication Union (ITU) and is based in Geneva, Switzerland. The various principles of ISDN as per ITU-T recommendation are:

- To support switched and non-switched applications
- To support voice and non-voice applications
- Reliance on 64-kbps connections
- Intelligence in the network
- Layered protocol architecture
- Variety of configurations

Advantages of ISDN

- ISDN channels have a reliable connection.
- ISDN is used to facilitate the user with multiple digital channels.
- It has faster data transfer rate.
- Efficient use of bandwidth
- Improved call quality
- It provides Greater flexibility
- Integrated services

Disadvantages of ISDN

- ISDN lines costlier than the other telephone system.
- It requires specialized digital devices.
- It is less flexible.
- Limited coverage
- High installation and maintenance costs
- Limited features

UNIT–2 : DATA LINK LAYER 🚀

Data Link Layer Design Issues Data-link layer is the second layer after the physical layer. The data link layer is responsible for maintaining the data link between two hosts or nodes.

1)Logical Link Control Sub-layer (LLC) -

2)Media Access Control Sub-layer (MAC)

Main Functions

- **Framing**
- **Error detection**
- **Error correction**
- **Flow control**
- **Access control**

1. Framing

The network layer gives a stream of bits to the Data Link Layer.
The Data Link Layer divides this data into small units called frames.

. Byte Stuffing

2. Flow Control

Flow control manages the speed of transmission between sender and receiver.

Techniques

- Stop and Wait
- Sliding Window

3. Error Control

Errors may occur during transmission because of noise.

Techniques

- Parity Check
- Checksum
- CRC

4. Access Control

When many devices share the same channel, access control decides which device can send data.

Protocols

- ALOHA
- CSMA/CD
- CSMA/CA

Framing in Data Link Layer

Framing is the process of dividing a stream of data into smaller units called frames.

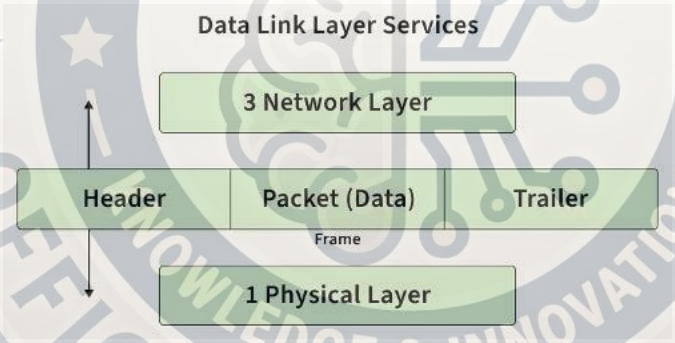
The Data Link Layer receives data from the Network Layer and converts it into frames for transmission.

Need of Framing

- Easy transmission
- Error checking
- Data organization

Structure of a Frame

Field	Purpose
Header	Address and control information
Data	Actual message
Trailer	Error checking information



Types of Framing

1. Character Count Method

The first field of the frame tells the number of characters in the frame.

2. Byte Stuffing

Special bytes are added at the beginning and end of the frame.

If the same byte appears inside data, an extra escape character is inserted.

3. Bit Stuffing

Used in bit-oriented protocols.

Whenever five consecutive 1s occur, the sender inserts a 0 bit.

The receiver removes the extra 0.

Example of Bit Stuffing

Original Data:
111110111111

After Bit Stuffing:
111110011111101

Advantages of Framing

- **Reliable communication**
- **Easy error detection**
- **Efficient data transfer**

Disadvantages of Framing

- **Extra overhead**
- **Complex processing in some methods**

Error Detection in Computer Networks

When data travels from sender to receiver, some bits may change because of:

- **Noise**
- **Interference**
- **Weak signal**
- **Hardware problems**

Error Detection techniques help the receiver check whether the received data is correct or corrupted.

Example of Error

Original Data:
1011001

Received Data:
1010001

Here one bit changed during transmission.

Types of Errors

1. Single Bit Error

Only one bit changes.

Example

Original:
101101

Received:
101001

2. Burst Error

Many bits change together.

Example

Original:
11001100

Received:
10011110

Burst errors are more common in real networks.

Error Detection Techniques

- Parity Check
- Checksum
- CRC

EVEN PARITY EXAMPLE

Question 1

Add even parity bit to:

1011001

Solution

Step 1: Count number of 1s

1011001

Number of 1s = 4

4 is already EVEN.

So parity bit = 0

Final Data

10110010

Question 2

Add even parity bit to:

110101

Solution

Count 1s:

110101

Number of 1s = 4

Already even.

Parity bit = 0

Final Data:
1101010

Question 3

Add odd parity bit to:

101101

Solution

Count 1s:

101101

Number of 1s = 4

For ODD parity total must be odd.

So add parity bit = 1

Final Data:
1011011

1)ERROR DETECTION USING PARITY

Question 4

Received data:

1011011

Check using even parity.

Solution

Count number of 1s:

1011011

Total 1s = 5

But even parity requires EVEN number of 1s.

Hence:

ERROR DETECTED

Advantages of Parity Check

- **Very simple**
- **Fast**
- **Low cost**

Disadvantages

- **Cannot detect multiple errors**
- **Weak security**

2. CHECKSUM

Its adds binary data blocks.

Sender and receiver compare results.

Working Steps

1. Divide data into equal blocks

2. Add blocks
3. Send sum as checksum
4. Receiver verifies checksum

CHECKSUM EXAMPLE

Question 5

Find checksum for:

1010
1100

Using 4-bit checksum.

Solution

Step 1: Add binary numbers

1010
+ 1100

10110

Since result is 5 bits:

Carry = 1

Remaining bits = 0110

Add carry:

0110 + 1 = 0111

Step 2: Take 1’s Complement

0111 → 1000

Final Checksum

1000

Receiver Side Verification

Data:
1010
1100
Checksum:
1000

Add all:

1010
1100
1000

11110

Carry = 1

1110 + 1 = 1111

Complement of 1111 = 0000

Hence:

NO ERROR

Advantages of Checksum

- **Better than parity**
- **Used in TCP/IP**

Disadvantages

- **Less powerful than CRC**

3. CRC (Cyclic Redundancy Check)

CRC is the most powerful error detection technique.

Used in:

- Ethernet
- Wi-Fi
- Networks

CRC IDEA

CRC uses:

- Binary division
- Generator polynomial

CRC STEPS

1. Append zeros to data
2. Divide by generator
3. Get remainder
4. Append remainder to data
5. Receiver repeats division

If remainder = 0 → No error

CRC EXAMPLE

Question 6

Dataword = 1101
Generator = 1011

Find CRC.

Solution

Generator length = 4

Append 3 zeros:

1101000

Now perform XOR division.

Step-by-Step XOR Division

1101 ⊕ 1011 = 0110

Bring next bit:

1100

1100 ⊕ 1011 = 0111

Bring next bit:

1110

1110 ⊕ 1011 = 0101

Bring next bit:

1010

1010 ⊕ 1011 = 0001

Remainder = 001

Final Codeword

Original data:
1101

CRC:
001

Final transmitted data:

1101001

Advantages of CRC

- **Detects burst errors**
- **Highly reliable**
- **Used in real networks**

Disadvantages

- **Complex calculations**

COMPARISON OF TECHNIQUES

Technique	Accuracy	Complexity
Parity Check	Low	Simple
Checksum	Medium	Moderate
CRC	High	Complex

Error Correction in Computer Networks

Error correction is the process of detecting and correcting errors in transmitted data so that the receiver gets the original correct data without retransmission.

errors may occur because of:

- **Noise**
- **Signal interference**

- **Weak communication signals**
- **Hardware faults**

Sometimes detecting the error is not enough.
The receiver must also correct the error automatically.

This process is called Error Correction.

Need for Error Correction

1. **It improves reliability of communication.**
2. **It reduces retransmission of data.**
3. **It saves bandwidth and time.**
4. **It is useful in noisy communication channels.**
5. **It ensures accurate data delivery.**

Difference Between Error Detection and Error Correction

Error Detection	Error Correction
Detects errors only	Detects and corrects errors
May require retransmission	Corrects automatically
Simpler	More complex
Example: CRC	Example: Hamming Code

Types of Error Correction

There are two main methods:

1. **Forward Error Correction (FEC)**
2. **Automatic Repeat Request (ARQ)**

1. Forward Error Correction (FEC)

In FEC, the sender adds extra redundant bits to the data.

The receiver uses these bits to detect and correct errors automatically.

Features of FEC

- No retransmission needed
- Fast communication
- Useful in wireless and satellite communication

Advantages of FEC

1. Fast error correction
2. No need for retransmission
3. Useful in real-time systems

Disadvantages of FEC

1. Complex implementation
2. Extra redundant bits increase overhead

Example of FEC

- **Hamming Code**
- **Reed-Solomon Code**

2)1. Stop and Wait ARQ

The sender sends one frame and waits for acknowledgment.

If ACK is not received, the frame is retransmitted**.**

Working

1. **Send frame**

2. **Wait for ACK**
3. **If ACK received → send next frame**
4. **If timeout occurs → resend frame**

Advantages

- **Simple implementation**

Disadvantages

- **Slow communication**

2)2. Go-Back-N ARQ

The sender can send multiple frames continuously.

If one frame is damaged, all frames after that frame are retransmitted.

Example

Frames Sent:
1 2 3 4 5

Suppose frame 3 is damaged.

Then frames 3, 4, and 5 are resent.

Advantages

- Faster than Stop and Wait

Disadvantages

- Wastes bandwidth

2)3. Selective Repeat ARQ

Only the damaged frame is retransmitted.

Example

Frames Sent:
1 2 3 4 5

If frame 3 is damaged, only frame 3 is resent.

Advantages

- **Efficient**
- **Saves bandwidth**

Disadvantages

- **Complex implementation**

Hamming Code

Hamming Code is the most common error correction technique.

It can:

- Detect errors
- Correct single-bit errors

Working of Hamming Code

1. Redundant parity bits are added.
2. Receiver checks parity positions.
3. Error position is identified.
4. The incorrect bit is corrected.

Hamming Code Example

Question

Generate Hamming code for data:

1011

Step 1: Determine Number of Parity Bits

Formula:

$2^r \geq m + r + 1$

Where:

- r = parity bits
- m = data bits

For m = 4:

Try r = 3

$2^3 = 8$
$m + r + 1 = 4 + 3 + 1 = 8$

So, 3 parity bits are required.

Step 2: Position the Bits

Parity bits are placed at positions:
1, 2, 4

Data bits are placed at:
3, 5, 6, 7

Step 3: Fill Data Bits

Data = 1011

Position	Value
1	P1
2	P2
3	1
4	P4
5	0
6	1
7	1

Step 4: Calculate Parity Bits

P1 checks positions:

1, 3, 5, 7

Values:
1, 0, 1

Number of 1s = 2 (even)

P1 = 0

P2 checks positions:

2, 3, 6, 7

Values:
1, 1, 1

Number of 1s = 3 (odd)

P2 = 1

P4 checks positions:

4, 5, 6, 7

Values:
0, 1, 1

Number of 1s = 2 (even)

P4 = 0

Final Hamming Code

0110011

Advantages of Hamming Code

1. **Corrects single-bit errors**
2. **Reliable communication**
3. **Automatic correction**

Disadvantages of Hamming Code

1. **Complex for large data**
2. **More redundant bits required**

Applications of Error Correction

- Wi-Fi communication
- Satellite communication
- Mobile networks
- Data storage devices
- Deep-space communication

HDLC (High-Level Data Link Control)

HDLC is a bit-oriented Data Link Layer protocol developed by ISO for reliable communication between devices.

It provides:

- Error control
- Flow control
- Reliable data transmission

HDLC is mainly used in WAN (Wide Area Network) communication and supports both point-to-point and multipoint connections.

PPP (Point-to-Point Protocol)

PPP is a byte-oriented Data Link Layer protocol used for direct communication between two devices.

It is mainly used in:

- Internet connections
- Dial-up connections

PPP provides:

- Error detection
- Authentication

- Support for multiple network layer protocols

Difference Between HDLC and PPP

HDLC	PPP
HDLC stands for High-Level Data Link Control.	PPP stands for Point-to-Point Protocol.
It is a bit-oriented protocol.	It is a byte-oriented protocol.
Developed by ISO.	Developed as an Internet standard.
Supports point-to-point and multipoint communication.	Supports only point-to-point communication.
More complex protocol.	Simpler protocol.
.	
Uses bit stuffing.	Uses byte stuffing.
Mainly used in WAN communication.	Mainly used for Internet communication.
Cisco routers use HDLC as default protocol.	PPP is widely used in dial-up and Internet connections.
Less flexible.	More flexible and supports multiple protocols.

Detailed Difference Table

Feature	Stop and Wait	Go-Back-N	Selective Repeat

1)frames sent	One at a time	Multiple	Multiple
2)ACK	After every frame	Cumulative ACK	Individual ACK
3)Retransmission	One frame	All frames after error	Only damaged frame
4)Speed	Slow	Fast	Fastest
5)Complexity	Simple	Moderate	Complex
6)Bandwidth utilization	Poor	Better	Best
7)Memory requirement	Low	Moderate	High
8)Efficiency	Low	Medium	High
9)Window size	1	Multiple	Multiple
10)Buffering	Not needed	Small buffering	Large buffering

Difference Between Stop and Wait Protocol and Sliding Window Protocol

Stop and Wait Protocol	Sliding Window Protocol
Sender sends only one frame at a time.	Sender can send multiple frames at a time.
Sender waits for ACK after every frame.	Sender does not wait for ACK for every frame.
Window size is 1.	Window size is more than 1.
Communication speed is slow.	Communication speed is fast.
Low bandwidth utilization.	Better bandwidth utilization.
Simple and easy to implement.	More complex than Stop and Wait.
Less efficient for large networks.	More efficient for large networks.
No pipelining is used.	Uses pipelining technique.
Suitable for short-distance communication.	Suitable for long-distance communication.

Less memory required.

More memory required.

1. Sliding Window Protocol

Sliding Window Protocol is a flow control method in which the sender can send multiple frames continuously before receiving acknowledgments (ACKs).

It improves:

- Speed
- Efficiency
- Bandwidth utilization

Why We Need Sliding Window?

In Stop and Wait protocol:

Send one frame → wait → send next frame

This wastes time because the sender keeps waiting.

Sliding Window solves this problem by sending many frames together.

Example of Sliding Window

Suppose:

Window Size = 4

Frames:
1, 2, 3, 4, 5, 6, 7

Step 1

Sender sends first 4 frames:

1 2 3 4

Without waiting.

Step 2

Receiver sends ACK for Frame 1.

Now window slides forward.

Sender can now send Frame 5.

2 3 4 5

Step 3

ACK for Frame 2 received.

Sender sends Frame 6.

3 4 5 6

Window continuously moves forward.

This movement is called:
👉 Sliding Window

Advantages of Sliding Window

1. **Faster communication**
2. **Better bandwidth utilization**
3. **Continuous transmission**
4. **Efficient for large networks**

Disadvantages

1. **More complex**

2. **Requires buffer memory**

Types of Sliding Window Protocol

1. **Go-Back-N ARQ**
2. **Selective Repeat ARQ**

2. Go-Back-N ARQ

Go-Back-N ARQ is a sliding window protocol in which the sender can send multiple frames continuously.

If one frame is lost or damaged, that frame and all following frames are retransmitted.

Why Called “Go-Back-N”?

Because sender goes back and retransmits multiple frames after the error frame.

Working of Go-Back-N

Suppose:

Window Size = 4

Frames:
1 2 3 4 5 6

Step 1 — Sender Sends Frames

1 2 3 4

Step 2 — Error Occurs

Suppose Frame 3 is damaged.

Receiver receives:

1 2 ❌ 4

Receiver accepts:

- 1
- 2

But rejects:

- 3
- 4

Step 3 — ACK Sent

Receiver sends ACK only up to Frame 2.

Step 4 — Sender Retransmits

Sender retransmits:

3 4

And then continues with:
5, 6

Important Point

Even if Frame 4 was correct, it is retransmitted again.

This wastes bandwidth.

Diagram of Go-Back-N

Sender → 1 → Receiver ✓
Sender → 2 → Receiver ✓
Sender → 3 → Receiver ✗

Sender → 4 → Receiver Rejected

Resend:
3 4

Features of Go-Back-N

- **Uses sliding window**
- **Multiple frames sent together**
- **Cumulative acknowledgment used**
- **Error recovery by retransmission**

Advantages of Go-Back-N

1. **Faster than Stop and Wait**
2. **Better bandwidth utilization**
3. **Continuous transmission possible**

Disadvantages of Go-Back-N

1. **Wastes bandwidth**
2. **Retransmits correct frames also**
3. **Less efficient than Selective Repeat**

MAC (Media Access Control)

MAC (Media Access Control) is a sublayer of the Data Link Layer that controls how multiple devices access and share a common communication channel.

It decides:

- Which device can send data
- When data can be sent
- How collisions are avoided

Position of MAC Layer in OSI Model

The Data Link Layer has two sublayers:

Data Link Layer
├── LLC (Logical Link Control)
└── MAC (Media Access Control)

Main Functions of MAC

1. Channel Access Control

MAC decides which device gets permission to use the communication channel.

2. Collision Handling

When two devices send data at the same time, collision occurs.

MAC helps:

- Detect collisions
- Avoid collisions

3. Physical Addressing

MAC uses MAC addresses for communication inside a LAN.

Example of MAC address:

00:1A:2B:3C:4D:5E

4. Frame Transmission

MAC controls frame sending and receiving between devices.

MAC Address

Every network device has a unique MAC address.

It is:

- **48 bits long**
- **Hardware address**
- **Permanently assigned**

MAC Address Format

AA:BB:CC:DD:EE:FF

Types of MAC Protocols

1. Random Access Protocols;-Devices send data randomly.

Examples:

- ALOHA
- CSMA
- CSMA/CD
- CSMA/CA

2. Controlled Access Protocols

Devices get permission before transmission.

Examples:

- **Token Passing**
- **Polling**

3. Channelization Protocols

Channel divided among users.

Examples:

- **FDMA**
- **TDMA**
- **CDMA**

Advantages of MAC

1. **Reduces collisions**
2. **Improves network efficiency**
3. **Controls channel access**
4. **Supports reliable communication**

Disadvantages of MAC

1. **Complex management in large networks**
2. **Delay may occur during heavy traffic**

Applications of MAC

- **Ethernet**
- **Wi-Fi**
- **LAN communication**
- **Wireless networks**

1. ALOHA Protocol

ALOHA is a random access MAC(Media Access Control) protocol in which a device sends data whenever it wants without checking whether the communication channel is free or busy.

If two devices transmit data at the same time, collision occurs. After collision, the sender waits for a random amount of time and retransmits the data.

ALOHA was developed for wireless communication systems.

Working of ALOHA

1. **Device sends data immediately.**
2. **If acknowledgment (ACK) is received → transmission successful.**
3. **If collision occurs → no ACK received.**
4. **Sender waits random time and retransmits data.**

Types of ALOHA

(i) Pure ALOHA

In Pure ALOHA, data can be transmitted at any time.

Features

- **Very simple**
- **High collision probability**
- **Low efficiency**

(ii) Slotted ALOHA

In Slotted ALOHA, transmission is allowed only at fixed time slots.

Features

- **Fewer collisions**
- **Better efficiency than Pure ALOHA**

Advantages of ALOHA

1. Simple implementation
2. No synchronization required (Pure ALOHA)
3. Easy access method

Disadvantages of ALOHA

1. High collision rate
2. Low efficiency
3. Poor bandwidth utilization

Applications of ALOHA

- Satellite communication
- Wireless communication systems

2. CSMA (Carrier Sense Multiple Access)

CSMA is a MAC protocol in which a device first listens to the communication channel before sending data.

If the channel is free, data is transmitted.
If the channel is busy, the device waits before transmitting.

The main purpose of CSMA is to reduce collisions.

Working of CSMA

1. Device senses the channel.
2. If channel is idle → send data.
3. If channel is busy → wait.
4. Retry transmission later.

Types of CSMA

(i) 1-Persistent CSMA

The device continuously senses the channel.

As soon as the channel becomes free, it immediately transmits data.

Advantage

- **Simple method**

Disadvantage

- **High collision probability**

(ii) Non-Persistent CSMA

If the channel is busy, the device waits for a random time before checking again.

Advantage

- **Fewer collisions**

Disadvantage

- **Delay increases**

(iii) P-Persistent CSMA

Used mainly in slotted channels.

The device transmits with probability P when the channel becomes free.

Advantages of CSMA

1. Reduces collisions
2. Better efficiency than ALOHA
3. Better bandwidth utilization

Disadvantages of CSMA

1. Collisions can still occur
2. Delay may happen in busy networks

Applications of CSMA

- Ethernet LANs
- Shared communication networks

3. CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

CSMA/CD is a MAC protocol used mainly in Ethernet networks in which a device senses the channel before transmission and also detects collisions during transmission.

If collision occurs, transmission stops immediately and data is retransmitted later.

Working of CSMA/CD

1. **Listen to the channel.**
2. **If channel is free → send data.**
3. **While sending, monitor collision.**
4. **If collision occurs → stop transmission.**
5. **Send jam signal.**
6. **Wait random time.**
7. **Retransmit data.**

Collision in CSMA/CD

Collision happens when two devices send data at the same time.

Advantages of CSMA/CD

1. **Reduces collision effects**
2. **Efficient communication**
3. **Better network performance**

Disadvantages of CSMA/CD

1. **Not suitable for wireless networks**
2. **Performance decreases in heavy traffic**

Applications of CSMA/CD

- **Ethernet networks**
- **Wired LAN communication**

4. CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

CSMA/CA is a MAC protocol mainly used in wireless networks (Wi-Fi) in which devices try to avoid collisions before transmitting data.

Instead of detecting collisions, it prevents them from occurring.

Why Collision Avoidance?

In wireless communication, collision detection is difficult because devices cannot transmit and listen at the same time.

Therefore, collisions are avoided before transmission.

Working of CSMA/CA

1. Device senses the channel.
2. If channel is idle → wait for a random backoff time.
3. Send RTS (Request to Send).
4. Receiver sends CTS (Clear to Send).

Advantages of CSMA/CA

1. Reduces collisions in wireless networks
2. Improves wireless communication reliability
3. Efficient for Wi-Fi networks

Disadvantages of CSMA/CA

1. **More delay because of waiting time**
2. **More overhead due to RTS/CTS**

Applications of CSMA/CA

- **Wi-Fi networks**
- **Wireless LANs**
- **IEEE 802.11**

Difference Between ALOHA, CSMA, CSMA/CD, and CSMA/CA

Protocol	Main Idea	Collision Handling	Used In
ALOHA	Send anytime	Collision after transmission	Wireless/Satellite
CSMA	Listen before send	Reduces collision	LAN
CSMA/CD	Detect collision	Detects collision after sending	Ethernet
CSMA/CA	Avoid collision	Avoids collision before sending	Wi-Fi

IEEE 802.3 (Ethernet)

IEEE 802.3 is the standard for Ethernet networks that uses CSMA/CD (Carrier Sense Multiple Access with Collision Detection) for communication in wired Local Area Networks (LANs).

It defines:

- Physical layer specifications
- Data Link layer specifications
- Frame format
- Transmission method

Features of IEEE 802.3

- Uses bus or star topology
- Wired communication
- Uses CSMA/CD
- High-speed LAN communication

Applications

- **Office networks**
- **Computer labs**
- **Ethernet LANs**

IEEE 802.4 (Token Bus)

Definition

IEEE 802.4 is a LAN standard that uses a token-passing mechanism over a bus topology.

Only the device holding the token can transmit data.

Features of IEEE 802.4

- **Uses bus topology**

- **Token-passing protocol**
- **Collision-free communication**

Advantages

- **No collision**
- **Reliable communication**

Disadvantages

- **Complex implementation**
- **Less popular**

IEEE 802.5 (Token Ring)

IEEE 802.5 is a LAN standard that uses token passing in a ring topology.

A special frame called a token circulates around the ring.
Only the device with the token can send data.

Features of IEEE 802.5

- **Uses ring topology**
- **Collision-free transmission**
- **Controlled access method**

Advantages

- **No collisions**
- **Efficient communication**

Disadvantages

- Expensive
- Difficult maintenance

IEEE 802.11 (Wi-Fi)

IEEE 802.11 is the standard for Wireless Local Area Networks (WLANs), commonly known as Wi-Fi.

It uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).

Features of IEEE 802.11

- Wireless communication
- Supports mobility
- Uses radio waves

Advantages

- No cables required
- Easy installation
- Portable communication

Disadvantages

- Lower security
- Signal interference

Applications

- Wi-Fi networks
- Mobile devices

- **Wireless Internet**

FDDI (Fiber Distributed Data Interface)

FDDI is a high-speed network standard that uses optical fiber and dual ring topology for data transmission.

It is mainly used in large networks requiring high reliability and speed.

Features of FDDI

- **Uses optical fiber**
- **High-speed communication**
- **Dual ring topology**
- **Token-passing method**

Advantages

- **Very high speed**
- **Reliable communication**
- **Fault tolerance**

Disadvantages

- **Expensive**
- **Complex setup**

Applications

- **Backbone networks**
- **Large organizations**

SONET/SDH

SONET (Synchronous Optical Network) and SDH (Synchronous Digital Hierarchy) are standards used for high-speed optical fiber communication.

They provide synchronized digital data transmission over optical networks.

SONET

SONET is mainly used in the United States and Canada.

SDH

SDH is the international version used worldwide.

Features of SONET/SDH

- **High-speed transmission**
- **Optical fiber communication**
- **Synchronous communication**
- **Reliable long-distance communication**

Advantages

- **Very high bandwidth**
- **Reliable transmission**
- **Supports long-distance communication**

Disadvantages

- **High installation cost**
- **Complex management**

Applications

- **Telecommunication networks**
- **Internet backbone**
- **Fiber optic communication**

Difference Between IEEE Standards

Standard	Technology	Topology	Access Method
802.3	Ethernet	Bus/Star	CSMA/CD
802.4	Token Bus	Bus	Token Passing
802.5	Token Ring	Ring	Token Passing
802.11	Wi-Fi	Wireless	CSMA/CA

SONET Definition

SONET (Synchronous Optical Network) is a high-speed optical fiber communication standard developed by ANSI for synchronous digital data transmission mainly in North America.

SDH Definition

SDH (Synchronous Digital Hierarchy) is an international high-speed optical fiber communication standard developed by ITU-T for synchronized digital transmission worldwide.

Difference Between SONET and SDH

SONET	SDH
SONET stands for Synchronous Optical Network.	SDH stands for Synchronous Digital Hierarchy.
Developed by ANSI (American National Standards Institute).	Developed by ITU-T (International Telecommunication Union).
Mainly used in the United States and Canada.	Used internationally all over the world.
Standard for optical fiber communication in North America.	International standard for optical fiber communication.
Basic transmission rate is STS-1 (51.84 Mbps).	Basic transmission rate is STM-1 (155.52 Mbps).
Uses STS (Synchronous Transport Signal) levels.	Uses STM (Synchronous Transport Module) levels.
Slightly less flexible.	More flexible and globally accepted.

Supports synchronous digital transmission.	**Also supports synchronous digital transmission.**
Compatible with SDH at higher levels.	**Compatible with SONET at higher levels.**

Unit 3: Network Layer

Design Issues in Network Layer

Introduction

The Network Layer is the third layer of the OSI Model.

Its main function is:

- Routing packets from source to destination
- Logical addressing
- Internetworking

While designing the Network Layer, several important issues must be considered to ensure reliable and efficient communication.

These are called **Design Issues in Network Layer**.

Definition

Design issues in the Network Layer are the important problems and functions considered while designing the Network Layer for efficient packet delivery and routing in computer networks.

Main Design Issues in Network Layer

1. Store and Forward Packet Switching
2. Services Provided to Transport Layer
3. Implementation of Connectionless Service
4. Implementation of Connection-Oriented Service

5. Routing
6. Congestion Control
7. Quality of Service (QoS)
8. Internetworking

1. Store and Forward Packet Switching

In this method, a router receives the complete packet, stores it temporarily, checks it, and then forwards it to the next router.

Working

1. Packet arrives at router
2. Router stores packet
3. Error checking performed
4. Packet forwarded

Example

Like a courier office receiving a package before sending it to the next city.

2. Services Provided to Transport Layer

The Network Layer provides services to the Transport Layer.

Types of Services

(i) Connectionless Service

- No dedicated path
- Each packet travels independently
- Faster communication

Example

Internet communication (IP)

(ii) Connection-Oriented Service

- Dedicated path established before transmission
- Reliable communication

Example

Virtual circuits

Difference

Connectionless	Connection-Oriented
No setup required	Setup required
Fast	Reliable
Packets independent	Fixed path used

3. Implementation of Connectionless Service

Implemented using:

- Datagram subnet

Each packet is routed independently.

Different packets may follow different paths.

Features

- No route setup
- Flexible routing
- Fast communication

4. Implementation of Connection-Oriented Service

Implemented using:

- Virtual circuit subnet

A fixed path is established before communication starts.

Phases

1. Connection setup
2. Data transfer
3. Connection release

Advantages

- Reliable communication
- Packets arrive in order

Disadvantages

- Setup delay

5. Routing

Routing is the process of selecting the best path for transmitting packets from source to destination.

Functions of Routing

- Path selection
- Packet forwarding
- Traffic management

Types of Routing

(i) Static Routing

Routes are fixed manually.

Advantages

- Simple

Disadvantages

- Not flexible

(ii) Dynamic Routing

Routes change automatically according to network conditions.

Advantages

- Flexible
- Efficient

Disadvantages

- Complex

Examples of Routing Protocols

- RIP
- OSPF
- BGP

6. Congestion Control

Congestion occurs when too many packets are present in the network and network performance becomes slow.

Congestion control techniques reduce traffic overload.

Causes of Congestion

- Heavy traffic
- Slow routers
- Insufficient bandwidth

Effects of Congestion

- Packet loss
- Delay
- Reduced performance

Congestion Control Methods

1. Traffic shaping
2. Queue management
3. Admission control

7. Quality of Service (QoS)

QoS ensures better network performance for important applications.

QoS Parameters

Parameter	Meaning
Bandwidth	Data transfer capacity
Delay	Transmission time
Jitter	Delay variation
Reliability	Accurate delivery

Applications of QoS

- Video conferencing
- Online gaming
- VoIP calls

8. Internetworking

Internetworking means connecting multiple networks together.

Routers are mainly used for internetworking.

Purpose

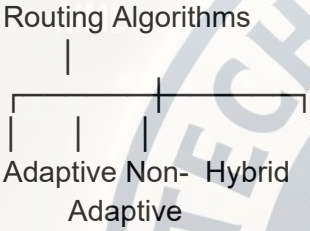
- Communication between different networks
- Global connectivity

Example

Internet is the best example of internetworking.

Classification of Routing Algorithms

Routing algorithms are mainly classified into:



1. Adaptive Routing Algorithms

Adaptive routing algorithms change their routing decisions according to:

- Network topology changes
- Traffic load
- Delay
- Congestion

These are also called:
👉 Dynamic Routing Algorithms

Features

- Routes change dynamically
- Uses current network information
- Better for large networks

Advantages

1. Flexible routing
2. Better performance
3. Handles congestion efficiently
4. Automatic rerouting possible

Disadvantages

1. Complex implementation
2. More processing overhead
3. Higher memory usage

Types of Adaptive Routing Algorithms

(i) Isolated Adaptive Algorithm

Each node makes routing decisions using only its own local information.

It does not take information from neighboring routers.

Examples

- Hot Potato Routing
- Backward Learning

(ii) Centralized Adaptive Algorithm

A central node maintains complete information about the network and makes routing decisions for all routers.

Example

👉 Link State Routing

(iii) Distributed Adaptive Algorithm

Each router receives information from neighboring routers and independently calculates the best path.

Also called:
👉 Decentralized Routing

Example

👉 Distance Vector Routing

2. Non-Adaptive Routing Algorithms

Non-adaptive routing algorithms do not change routing decisions after routes are selected.

These are also called:
👉 Static Routing Algorithms

Features

- Fixed routes
- Manual configuration
- No automatic updates

Advantages

1. Simple implementation
2. Low overhead
3. Easy management in small networks

Disadvantages

1. Not flexible
2. Cannot handle failures automatically
3. Poor for large networks

Types of Non-Adaptive Routing Algorithms

(i) Flooding

Definition

In flooding, every incoming packet is sent through all outgoing links except the one from which it arrived.

Problems in Flooding

- Packet looping
- Duplicate packets

Solutions

1. Sequence Number
2. Hop Count
3. Spanning Tree

Real-Life Example 😄

Like forwarding a message to everyone in your contact list.

(ii) Random Walk

In Random Walk routing, packets are forwarded randomly from one node to another neighbor.

3. Hybrid Routing Algorithms

Hybrid routing algorithms combine both adaptive and non-adaptive routing methods.

The network is divided into regions and different algorithms are used in different regions.

Features

- Combination approach
- Better efficiency
- Flexible routing

Types of Hybrid Routing

(i) Link-State Routing

Definition

Each router creates a complete map of the network and shares it with other routers.

Features

- Full topology information
- Fast routing decisions

Advantages

1. Fast convergence
2. Accurate routing
3. No routing loops

Disadvantages

1. Complex implementation
2. More memory required

Example

👉 OSPF

(ii) Distance Vector Routing

Definition

Each router maintains a routing table containing:

- Distance
- Direction

to every other router.

Routers share this table with neighbors.

Features

- Neighbor-based routing
- Uses Bellman-Ford algorithm

Advantages

1. Simple implementation
2. Low complexity

Disadvantages

1. Routing loops possible
2. Slow convergence

Example

👉 RIP

Difference Between Adaptive and Non-Adaptive Routing

Adaptive Routing	Non-Adaptive Routing
Routes change dynamically	Routes remain fixed
Uses current network conditions	No network condition updates
Complex	Simple
Suitable for large networks	Suitable for small networks
Better performance	Less flexible

Congestion Control Algorithms

Definition

Congestion control algorithms are techniques used in computer networks to control network traffic and prevent congestion.

These algorithms ensure that the network does not become overloaded with too many packets.

What is Congestion?

Congestion occurs when:

- Too many packets enter the network
- Routers cannot handle traffic properly
- Network becomes slow

As a result:

- Delay increases
- Packets are lost
- Performance decreases

Need for Congestion Control

Congestion control is important because it:

1. Prevents network overload
2. Reduces packet loss
3. Improves network performance
4. Controls traffic flow
5. Increases efficiency

Causes of Congestion

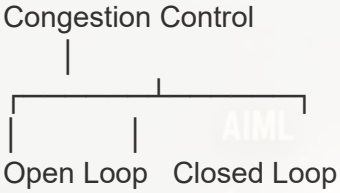
1. Heavy network traffic
2. Slow routers
3. Limited bandwidth
4. Large number of users
5. High-speed sender with slow receiver

Effects of Congestion

1. Packet loss
2. Delay in transmission
3. Reduced throughput
4. Retransmission of packets
5. Network failure in severe cases

Types of Congestion Control Algorithms

Congestion control algorithms are mainly classified into:



---

1. Open Loop Congestion Control

Definition

Open loop congestion control tries to prevent congestion before it occurs.

No feedback from the network is used.

Features

- Prevention-based method
- No monitoring of congestion
- Simple implementation

Techniques of Open Loop Control

(i) Retransmission Policy

Controls unnecessary retransmissions.

(ii) Window Policy

Controls number of packets sent.

(iii) Acknowledgment Policy

Reduces unnecessary acknowledgments.

(iv) Discarding Policy

Routers discard less important packets during congestion.

(v) Admission Policy

Controls entry of new traffic into the network.

Advantages

1. Simple
2. Less overhead

Disadvantages

1. Cannot react to actual congestion

---

2. Closed Loop Congestion Control

Definition

Closed loop congestion control detects congestion and takes corrective actions using feedback from the network.

Features

- Feedback-based method
- Monitors network conditions
- More efficient

Techniques of Closed Loop Control

(i) Backpressure

Congested node informs previous nodes to slow down transmission.

(ii) Choke Packet

Router sends a choke packet to sender asking to reduce transmission rate.

(iii) Implicit Signaling

Sender detects congestion by observing:

- Delay
- Packet loss

(iv) Explicit Signaling

Routers directly inform sender about congestion.

Advantages

1. Better congestion handling
2. Dynamic traffic control

Disadvantages

1. More complex
2. Extra overhead

Important Congestion Control Algorithms

---

1. Leaky Bucket Algorithm

Definition

Leaky Bucket algorithm controls traffic by sending packets at a constant rate.

Working

- Incoming packets stored in bucket
- Packets leak out at fixed speed
- Extra packets discarded if bucket becomes full

Features

- Smooth traffic flow
- No burst traffic allowed

Advantages

1. Prevents sudden traffic burst
2. Controls transmission rate

Disadvantages

1. Burst traffic not supported

Real-Life Example 😄

Like water leaking from a bucket at fixed speed.

---

2. Token Bucket Algorithm

Definition

Token Bucket algorithm allows burst traffic while controlling average transmission rate.

Working

- Tokens generated continuously
- To send packet, token required
- If enough tokens available → burst transmission possible

Features

- Supports burst traffic
- Flexible transmission

Advantages

1. Allows burst traffic
2. Better bandwidth usage

Disadvantages

1. More complex than Leaky Bucket

Real-Life Example 😄

Like collecting tokens and spending them together.

Difference Between Leaky Bucket and Token Bucket

Leaky Bucket	Token Bucket
Fixed output rate	Variable output rate
Burst traffic not allowed	Burst traffic allowed
Simpler	More flexible
Packets discarded if bucket full	Tokens control transmission

Congestion Control vs Flow Control

Congestion Control	Flow Control
Controls network traffic	Controls sender-receiver speed
Prevents network overload	Prevents receiver overload
Works at network level	Works between sender and receiver

Quality of Service (QoS)

Definition

Quality of Service (QoS) is a network technique used to provide better and guaranteed performance for important applications and network traffic.

QoS manages:

- Bandwidth
- Delay
- Packet loss
- Traffic priority

to ensure reliable communication.

Need for QoS

Different applications need different network performance.

For example:

- Video calls need low delay
- File transfer needs high bandwidth

QoS helps provide proper service according to application requirements.

Goals of QoS

1. Improve network performance
2. Reduce delay
3. Reduce packet loss
4. Control congestion
5. Prioritize important traffic

Main QoS Parameters

---

1. Bandwidth

Definition

Bandwidth is the amount of data that can be transmitted through a network in a given time.

Measured in:

- bps
- Mbps
- Gbps

Importance

Higher bandwidth → Faster data transfer

Example

Video streaming requires high bandwidth.

2. Delay (Latency)

Definition

Delay is the time taken for a packet to travel from source to destination.

Types of Delay

1. Transmission Delay
2. Propagation Delay
3. Processing Delay
4. Queuing Delay

Importance

Low delay is important for:

- Online gaming
- Video calls

---

3. Jitter

Definition

Jitter is the variation in packet delay during transmission.

Importance

High jitter causes:

- Voice breaking
- Video buffering

Example

In video calls, uneven packet arrival creates disturbance.

---

4. Reliability

Definition

Reliability means accurate and successful delivery of packets without errors.

Importance

Important for:

- Banking systems
- Online transactions

---

5. Throughput

Definition

Throughput is the actual amount of data successfully transmitted per unit time.

Importance

Higher throughput improves network efficiency.

QoS Techniques

1. Traffic Shaping

Controls data transmission rate.

Types

Technique	Work
Leaky Bucket	Fixed output rate
Token Bucket	Allows burst traffic

2. Packet Scheduling

Determines packet transmission order.

Methods

Method	Work
FIFO	First packet first
Priority Queuing	High-priority packets first
Round Robin	Equal turn for all

3. Resource Reservation

Reserves bandwidth for important applications.

---

4. Admission Control

Controls entry of new traffic to avoid overload.

QoS Service Models

---

1. Best Effort Service

- No guarantee
- All traffic treated equally

Example:
👉 Normal Internet service

---

2. Integrated Services (IntServ)

- Reserves resources
- Guaranteed QoS

---

3. Differentiated Services (DiffServ)

- Different priority levels
- Scalable QoS

Applications of QoS

1. Video conferencing
2. VoIP calls
3. Online gaming
4. Live streaming
5. Cloud computing
6. Banking systems

Advantages of QoS

1. Better network performance
2. Reduced delay and packet loss
3. Priority to important traffic
4. Improved user experience

Disadvantages of QoS

1. Complex implementation
2. Higher cost
3. Requires proper configuration

Real-Life Example 😄

Imagine a road with VIP lanes.

Emergency vehicles get priority.

Similarly, QoS gives priority to important network traffic.

Difference Between Bandwidth and Throughput

Bandwidth	Throughput

Maximum capacity	Actual transmitted data
Theoretical speed	Real speed

Difference Between Delay and Jitter

Delay	Jitter
Time taken by packet	Variation in delay
Constant	Changes continuously

Internetworking

Definition

Internetworking is the process of connecting two or more different computer networks so that they can communicate and exchange data with each other.

The connected networks together form an internetwork.

The biggest example of internetworking is:
👉 The Internet

Simple Meaning 😄

When multiple networks are connected together to share information, it is called internetworking.

Need for Internetworking

Internetworking is needed to:

1. Connect different networks

2. Share resources and data
3. Enable communication between distant locations
4. Increase network coverage
5. Improve data exchange

Main Functions of Internetworking

1. Communication between networks
2. Data routing
3. Packet forwarding
4. Address translation
5. Error handling

Devices Used in Internetworking

Device	Function
Router	Connects different networks
Gateway	Connects networks using different protocols
Bridge	Connects similar networks
Switch	Connects devices within a network

Router in Internetworking

Definition

A router is a Network Layer device that connects multiple networks and forwards packets between them.

Functions of Router

1. Path selection
2. Packet forwarding
3. Traffic management

Gateway in Internetworking

Definition

A gateway is a device that connects networks using different protocols.

It acts as a translator between networks.

Example

Connecting:

- TCP/IP network
- AppleTalk network

Types of Internetworking

---

1. LAN (Local Area Network)

Small geographical area.

Example:

- School
- Office

---

2. MAN (Metropolitan Area Network)

Covers a city or metropolitan area.

---

3. WAN (Wide Area Network)

Large geographical area.

Example:
👉 Internet

Components of Internetworking

1. Networks
2. Routers
3. Protocols
4. Transmission media

Protocols Used in Internetworking

Protocol	Work
IP	Logical addressing
ICMP	Error reporting
ARP	Address resolution
TCP	Reliable communication

Working of Internetworking

1. Sender sends packet
2. Router checks destination address
3. Router selects best path
4. Packet forwarded to destination network
5. Receiver receives packet

Advantages of Internetworking

1. Global communication
2. Resource sharing
3. High scalability
4. Better connectivity
5. Faster information exchange

Disadvantages of Internetworking

1. Security issues
2. Complex management
3. High maintenance cost
4. Congestion problems

The Network Layer in the Internet

Introduction

The Network Layer is the third layer of the OSI Model and an important layer in the Internet architecture.

Its main function is to transfer packets from the source network to the destination network through routers.

In the Internet, the Network Layer mainly uses:

- IP (Internet Protocol)
- ICMP
- IGMP
- Routing protocols like OSPF and BGP

Definition

The Network Layer in the Internet is responsible for logical addressing, routing, packet forwarding, and internetworking between different networks.

Main Functions of Network Layer in Internet

1. Logical Addressing
2. Routing
3. Packet Forwarding
4. Fragmentation
5. Internetworking
6. Congestion Control

IPv4 Address

Definition

IPv4 (Internet Protocol version 4) is a 32-bit logical addressing scheme used to identify devices on a network.

It is the most widely used IP version in computer networks.

Format of IPv4 Address

IPv4 address is written in **decimal format** and divided into 4 parts (octets):

192.168.1.1

Each part ranges from **0 to 255**.

Structure of IPv4

- Total bits: 32 bits
- Divided into 4 octets (8 bits each)
- Written in dotted decimal notation

Example:

11000000.10101000.00000001.00000001
= 192.168.1.1

Classes of IPv4 Addresses

Class	Range	Use
A	1–126	Large networks
B	128–191	Medium networks
C	192–223	Small networks
D	224–239	Multicasting
E	240–255	Research

Features of IPv4

- 32-bit address space
- Approximately 4.3 billion addresses
- Easy to use and implement
- Supports unicast, broadcast, and multicast

Advantages of IPv4

1. Simple addressing system
2. Widely used and supported
3. Easy routing

Disadvantages of IPv4

1. Limited address space
2. Security is not built-in
3. Exhaustion of IP addresses

Real-Life Example

IPv4 is like a house address used to identify a device on the internet.

IPv6 Address

Definition

IPv6 (Internet Protocol version 6) is a 128-bit addressing system designed to replace IPv4 and solve the problem of limited IP addresses.

Format of IPv6 Address

IPv6 is written in **hexadecimal format** and separated by colons:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Structure of IPv6

- Total bits: 128 bits
- Divided into 8 groups
- Each group has 16 bits
- Written in hexadecimal format

Simplified IPv6 Example

2001:db8::1

(Zero compression is allowed)

Features of IPv6

- 128-bit address space

- Huge number of IP addresses
- Built-in security (IPSec)
- Faster routing
- No need for NAT

Advantages of IPv6

1. Extremely large address space
2. Better security
3. Faster and efficient routing
4. No address exhaustion problem

Disadvantages of IPv6

1. Complex to implement
2. Requires upgrades in systems
3. Not fully adopted everywhere

Real-Life Example

IPv6 is like having unlimited house addresses for every device in the world.

Difference Between IPv4 and IPv6

Feature	IPv4	IPv6
Address size	32-bit	128-bit
Format	Decimal	Hexadecimal
Number of addresses	~4.3 billion	Very large (2^{128})
Security	Optional	Built-in (IPSec)
Header size	Smaller	Larger but efficient
Configuration	Manual/DHCP	Auto configuration

Example	192.168.1.1	2001:db8::1

Internet Control Protocols and Routing Protocols

These are important **Network Layer protocols used in the Internet** for addressing, routing, error control, and communication management.

---

1. IP (Internet Protocol)

Definition

IP is the main Network Layer protocol that is responsible for **logical addressing and packet delivery** across networks.

It ensures that data packets reach the correct destination using IP addresses.

Functions of IP

1. Logical addressing (IPv4 / IPv6)
2. Routing of packets
3. Fragmentation and reassembly
4. Best-effort delivery (no guarantee of delivery)

Types of IP

- IPv4 (32-bit)
- IPv6 (128-bit)

Key Point

👉 IP does not guarantee delivery; it only tries to deliver packets.

2. ICMP (Internet Control Message Protocol)

Definition

ICMP is a protocol used for **error reporting and network diagnostics** in IP networks.

Functions of ICMP

1. Reports errors in packet delivery
2. Helps in troubleshooting networks
3. Used for testing connectivity

Example

👉 `ping` command uses ICMP

ICMP Messages

Message	Purpose
Echo Request	Send test message
Echo Reply	Response to test
Destination Unreachable	Error message

Key Point

👉 ICMP helps in checking whether a device is reachable or not.

---

3. IGMP (Internet Group Management Protocol)

Definition

IGMP is a protocol used to manage **multicast groups** in IP networks.

Multicasting means sending data to multiple receivers at once.

Functions of IGMP

1. Manages group membership
2. Supports multicast communication
3. Adds/removes hosts from multicast groups

Applications

- Live video streaming
- Online broadcasting
- IPTV services

Key Point

👉 IGMP is used when one sender sends data to many receivers.

---

4. OSPF (Open Shortest Path First)

Definition

OSPF is a **link-state routing protocol** used to find the shortest path in an IP network.

It uses the **Dijkstra shortest path algorithm**.

Features of OSPF

- Fast routing
- Scalable for large networks
- Uses cost as metric
- Maintains full network topology

Working

1. Routers exchange link-state information
2. Build complete network map
3. Apply Dijkstra algorithm
4. Find shortest path

Advantages

1. Fast convergence
2. Efficient routing
3. No routing loops

Key Point

👉 OSPF is used inside large organizations (intra-domain routing).

---

5. BGP (Border Gateway Protocol)

Definition

BGP is a **path vector routing protocol** used to exchange routing information between different autonomous systems (AS) on the Internet.

It is the backbone of the Internet.

Features of BGP

- Used for inter-domain routing
- Maintains full path information
- Highly scalable
- Uses policy-based routing

Working

1. Exchanges routing information between AS
2. Selects best path based on policies
3. Maintains routing tables

Advantages

1. Internet-wide scalability
2. Reliable routing
3. Loop prevention using path information

Key Point

👉 BGP connects different networks of the Internet globally.

Difference Between OSPF and BGP

OSPF	BGP
Intra-domain routing	Inter-domain routing
Link-state protocol	Path vector protocol
Uses Dijkstra algorithm	Uses policy-based routing
Fast convergence	Slower but scalable
Used within organization	Used across Internet

Quick Summary Table

Protocol	Full Form	Function
IP	Internet Protocol	Addressing & routing
ICMP	Internet Control Message Protocol	Error reporting
IGMP	Internet Group Management Protocol	Multicast management
OSPF	Open Shortest Path First	Shortest path routing
BGP	Border Gateway Protocol	Internet routing between AS

Unit 4: Transport and Application Layer

The Transport Service (Computer Networks)

Introduction

The Transport Service is provided by the **Transport Layer (Layer 4 of OSI model)**. It is responsible for **end-to-end communication between processes running on different hosts**.

👉 In simple words: It ensures that data sent from one application reaches the correct application on another device.

Definition

The Transport Service is a set of services provided by the Transport Layer to enable **reliable, efficient, and error-free process-to-process communication** over a network.

Key Idea

- Network Layer → Host-to-Host delivery
- Transport Layer → Process-to-Process delivery

Main Services Provided by Transport Layer

---

1. Process-to-Process Delivery

It ensures that data is delivered to the correct application using **port numbers**.

Example:

- HTTP → Port 80
- FTP → Port 21

---

2. Segmentation and Reassembly

- Large messages are divided into smaller parts called **segments**
- At receiver side, segments are reassembled into original data

---

3. Connection Control

Transport service supports two types:

(i) Connection-Oriented Service (TCP)

- Connection is established before data transfer
- Reliable communication

(ii) Connectionless Service (UDP)

- No connection setup
- Faster but less reliable

---

4. Flow Control

Flow control ensures that:

👉 Sender does not overwhelm the receiver with too much data.

It balances the speed between sender and receiver.

---

5. Error Control

Ensures:

- Data is delivered correctly
- Lost or corrupted packets are retransmitted

---

6. Multiplexing and Demultiplexing

Multiplexing (Sender side)

Multiple applications send data using one network connection.

Demultiplexing (Receiver side)

Data is delivered to correct application using port numbers.

Transport Service Models

---

1. TCP Service (Reliable Service)

- Connection-oriented
- Reliable delivery
- Error detection and correction
- Ordered delivery

Example:
👉 Web browsing (HTTP), Email

---

2. UDP Service (Unreliable Service)

- Connectionless
- No guarantee of delivery
- Fast transmission

Example:
👉 Video streaming, Online gaming

Characteristics of Transport Service

1. End-to-end communication
2. Reliable or unreliable delivery
3. Flow control support
4. Error control
5. Port-based communication
6. Efficient data transfer

Transport Service vs Network Service

Transport Service	Network Service

Process-to-process	Host-to-host
Uses port numbers	Uses IP addresses
Provided by Transport Layer	Provided by Network Layer
Focus on applications	Focus on routing

Elements of Transport Protocols (Computer Networks)

Introduction

Transport protocols operate in the **Transport Layer (Layer 4 of OSI model)**. They are responsible for **end-to-end communication between processes**.

To perform this efficiently, transport protocols include several important components called **Elements of Transport Protocols**.

Definition

Elements of Transport Protocols are the basic functions and mechanisms used by the Transport Layer to ensure **reliable, ordered, and efficient delivery of data between applications**.

Main Elements of Transport Protocols

---

1. Addressing (Port Addressing)

Definition

Transport layer uses **port numbers** to identify the correct application on a device.

Types of Ports

Type	Range
Well-known ports	0–1023
Registered ports	1024–49151
Dynamic ports	49152–65535

Example

HTTP → Port 80
FTP → Port 21

Importance

- Ensures correct delivery to application
- Supports multiple applications on same device

---

2. Segmentation and Reassembly

Segmentation

Large data is divided into smaller units called **segments**.

Reassembly

At receiver side, segments are combined to form original message.

Example

A large file → broken into packets → sent → reassembled at destination.

Importance

- Efficient transmission
- Easier error handling

---

3. Connection Control

Defines how communication is established.

(i) Connection-Oriented Service (TCP)

- Connection is established before data transfer
- Reliable communication
- Uses handshake

(ii) Connectionless Service (UDP)

- No connection setup
- Faster communication
- Less reliable

Example

- TCP → Web browsing
- UDP → Video streaming

---

4. Flow Control

Definition

Flow control ensures that sender does not send data faster than receiver can handle.

Techniques

- Stop-and-Wait
- Sliding Window

Importance

- Prevents data loss
- Avoids receiver overload

---

5. Error Control

Definition

Error control ensures that data is delivered correctly without errors or loss.

Techniques

- Checksum
- Acknowledgment (ACK)
- Retransmission

Example

If packet is lost → sender retransmits it.

---

6. Multiplexing and Demultiplexing

Multiplexing (Sender Side)

Multiple applications share a single network connection.

Demultiplexing (Receiver Side)

Data is delivered to correct application using port numbers.

Example

One internet connection used for:

- WhatsApp
- YouTube
- Browser

---

7. Congestion Control

Definition

Congestion control prevents overload of network resources in transport communication.

Methods

- TCP Slow Start
- Window control
- Rate adjustment

Importance

- Reduces packet loss
- Improves performance

---

8. Reliable Delivery

Definition

Ensures that data is delivered correctly, in order, and without duplication.

Features

- Acknowledgment system
- Retransmission
- Sequence numbering

---

9. Ordering of Data

Definition

Ensures that packets are delivered in the correct order.

Example

Packet 1 → Packet 2 → Packet 3 (must arrive in same order)

Summary Table

Element	Function

Addressing	Identifies applications using ports
Segmentation	Divides data into small packets
Connection Control	Establishes or skips connection
Flow Control	Controls data speed
Error Control	Ensures correct delivery
Multiplexing	Combines multiple data streams
Congestion Control	Prevents overload
Reliable Delivery	Ensures accuracy
Ordering	Maintains sequence

Congestion Control in Transport and Application Layer

---

1. Introduction

Congestion control is a mechanism used in computer networks to **prevent overload of the network and ensure smooth data transmission**.

It is mainly handled by the **Transport Layer (TCP)**, but the **Application Layer also indirectly supports it** by controlling data generation and usage patterns.

---

2. What is Congestion?

Congestion occurs when:

- Too many packets enter the network
- Routers become overloaded
- Network delay and packet loss increase

---

3. Definition

Congestion control is the process of regulating the amount of data entering the network to avoid overload and maintain efficient communication.

---

4. Congestion Control in Transport Layer

The Transport Layer (mainly TCP) plays a **major role in congestion control**.

TCP Congestion Control

TCP uses feedback-based techniques to control congestion.

Key TCP Mechanisms

---

(1) Slow Start

- Starts with small congestion window (cwnd)
- Increases exponentially
- Quickly finds available bandwidth

---

(2) Congestion Avoidance

- After reaching threshold
- Increases window slowly (linear growth)

---

(3) Fast Retransmit

- If packet loss is detected (duplicate ACKs)
- Lost packet is retransmitted immediately

---

(4) Fast Recovery

- Instead of restarting from zero, TCP continues with reduced window

---

TCP Congestion Window (cwnd)

Controls how much data can be sent before receiving ACK.

Important Idea

👉 TCP continuously adjusts sending rate based on network condition.

---

5. Congestion Control in Application Layer

The Application Layer does not directly control congestion like TCP, but it **reduces congestion indirectly**.

How Application Layer Helps

---

(1) Rate Control

Applications reduce data generation rate.

Example:

- Video streaming adjusts quality (HD → SD)

---

(2) Buffering

- Data is stored temporarily before sending
- Prevents sudden traffic bursts

---

(3) Adaptive Streaming

- YouTube / Netflix adjust video quality based on network speed

---

(4) Traffic Reduction

- Compressing data before sending
- Reduces network load

---

6. Difference Between Transport and Application Layer Congestion Control

Feature	Transport Layer	Application Layer
Control Type	Direct control	Indirect control
Protocol Used	TCP	HTTP, Video apps, etc.
Mechanism	Window, ACK, retransmission	Compression, buffering, adaptation
Responsibility	Network-level control	User-level control
Accuracy	High precision	Approximate control

Transport Protocols: UDP and TCP

Transport protocols work in the **Transport Layer (Layer 4)** of the OSI model. They provide **process-to-process communication** between sender and receiver.

The two most important transport protocols are:

👉 **UDP (User Datagram Protocol)**
👉 **TCP (Transmission Control Protocol)**

---

1. UDP (User Datagram Protocol)

Definition

UDP is a **connectionless transport protocol** that sends data without establishing a connection between sender and receiver.

Features of UDP

- Connectionless protocol
- Fast communication
- No guarantee of delivery
- No ordering of packets
- No error recovery mechanism
- Low overhead

Working of UDP

1. Sender sends data directly
2. Packets are transmitted independently
3. Receiver gets packets without confirmation process

Advantages of UDP

1. Very fast
2. Simple protocol
3. Low overhead
4. Suitable for real-time applications

Disadvantages of UDP

1. Unreliable
2. Packet loss possible
3. No flow control
4. No error correction

Applications of UDP

- Video streaming
- Online gaming
- Voice calls (VoIP)
- DNS (Domain Name System)

Real-Life Example 😄

Like sending postcards:
👉 You send it directly without confirmation if it reached or not.

---

2. TCP (Transmission Control Protocol)

Definition

TCP is a **connection-oriented and reliable transport protocol** that ensures data is delivered correctly and in order.

Features of TCP

- Connection-oriented protocol
- Reliable communication
- Ordered delivery of data
- Error detection and correction
- Flow control and congestion control
- Higher overhead

TCP 3-Way Handshake

Before communication starts, TCP establishes a connection:

1. SYN → Client requests connection
2. SYN-ACK → Server responds
3. ACK → Client confirms

Working of TCP

1. Connection established
2. Data divided into segments
3. Sent with sequence numbers
4. Receiver sends ACK
5. Lost packets are retransmitted

Advantages of TCP

1. Reliable delivery
2. Ordered data transfer
3. Error control
4. Flow control
5. Congestion control

Disadvantages of TCP

1. Slower than UDP
2. More overhead
3. Complex protocol

Applications of TCP

- Web browsing (HTTP/HTTPS)
- Email (SMTP, POP3)
- File transfer (FTP)
- Secure communication

Real-Life Example 😄

Like sending registered mail:
👉 You get confirmation that delivery is successful.

Difference Between TCP and UDP

Feature	TCP	UDP
Type	Connection-oriented	Connectionless
Reliability	Reliable	Unreliable
Speed	Slower	Faster
Error Control	Yes	No
Flow Control	Yes	No
Ordering	Maintained	Not maintained
Overhead	High	Low
Example	Web, Email	Gaming, Streaming

Performance Problems in Computer Networks & Network Performance Measurement

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1. Performance Problems in Computer Networks

Introduction

Performance problems are issues that reduce the efficiency, speed, and reliability of a computer network. These problems occur when the network cannot handle data traffic properly.

Definition

Performance problems in computer networks are conditions that degrade network performance such as delay, packet loss, congestion, and low throughput.

Main Performance Problems

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1. Congestion

Definition

Congestion occurs when too many packets enter the network and exceed its capacity.

Effects

- Slow network speed
- Packet loss
- High delay

---

2. Delay (Latency)

Definition

Time taken for a packet to travel from source to destination.

Types

- Processing delay
- Transmission delay
- Propagation delay
- Queuing delay

Problem

High delay causes slow communication.

---

3. Packet Loss

Definition

When packets fail to reach the destination.

Causes

- Congestion
- Network errors
- Buffer overflow

---

4. Low Throughput

Definition

Actual amount of data successfully transmitted is low.

Problem

Network becomes inefficient.

---

5. Jitter

Definition

Variation in packet delay.

Problem

- Disturbs audio/video calls
- Causes buffering

---

6. Bandwidth Limitation

Definition

Limited capacity of network links.

Problem

Reduces speed of communication.

---

7. Network Failure

Definition

Complete breakdown of communication due to hardware/software failure.

Summary of Performance Problems

Problem	Effect
Congestion	Slow network

Delay	Late delivery
Packet Loss	Data missing
Jitter	Unstable communication
Low Throughput	Poor efficiency

Real-Life Example 😄

Imagine a road:

- Too many cars → congestion
- Traffic jam → delay
- Cars missing route → packet loss
- Uneven traffic → jitter

---

2. Network Performance Measurement

Introduction

Network performance measurement is used to evaluate how efficiently a network is working.

It helps in identifying network problems and improving performance.

Key Performance Metrics

---

1. Bandwidth

Definition

Maximum amount of data that can be transmitted in a network.

Unit

bps, Mbps, Gbps

Example

High bandwidth = faster internet

---

2. Throughput

Definition

Actual data successfully delivered in a network.

Difference

- Bandwidth = capacity
- Throughput = actual performance

---

3. Delay (Latency)

Definition

Time taken for data to travel from sender to receiver.

Formula idea

Includes transmission + propagation + processing + queuing delays

---

4. Jitter

Definition

Variation in delay of received packets.

Problem

Affects real-time applications like video calls.

---

5. Packet Loss

Definition

Percentage of packets lost during transmission.

Formula idea

(Packet sent - Packet received) / Packet sent × 100

---

6. Response Time

Definition

Time taken to get a response after sending a request.

---

7. Error Rate

Definition

Number of corrupted or damaged packets during transmission.

Tools for Network Measurement

1. Ping (checks connectivity)

2. Traceroute (finds path)
3. Throughput testers
4. Network analyzers

Application Layer (Computer Networks)

The Application Layer is the **topmost layer of the OSI model (Layer 7)**. It directly interacts with the user and provides network services to applications.

---

1. Introduction to Application Layer

Definition

The Application Layer is responsible for providing **network services directly to end-user applications** such as web browsers, email systems, file transfer tools, etc.

Main Functions

1. Provides user interface to network
2. Supports network services like web, email, file transfer
3. Enables communication between applications
4. Handles data representation and formatting

Examples

- Web browsing (Chrome, Firefox)
- Email (Gmail, Outlook)
- File transfer (FTP)

---

2. Client-Server Programming

Definition

Client-server programming is a network model where:

- **Client** requests services
- **Server** provides services

Working

1. Client sends request
2. Server processes request
3. Server sends response
4. Client receives data

Example

- Browser (Client) → Website (Server)
- Mobile app → Cloud server

Advantages

- Centralized control
- Easy management
- Efficient resource sharing

---

3. WWW (World Wide Web) and HTTP

WWW (World Wide Web)

Definition

WWW is a collection of interlinked web pages accessed through the Internet.

Features

- Uses hyperlinks
- Accessed via browsers
- Based on client-server model

HTTP (HyperText Transfer Protocol)

Definition

HTTP is a protocol used to transfer web pages between client and server.

Features

- Stateless protocol
- Works on TCP (Port 80)
- Fast communication

Working

1. Client sends HTTP request
2. Server processes request
3. Server sends web page response

Example

Typing `www.google.com` uses HTTP/HTTPS.

---

4. FTP (File Transfer Protocol)

Definition

FTP is a protocol used for transferring files between computers over a network.

Features

- Upload and download files
- Uses TCP
- Port 21

Modes

- Active mode
- Passive mode

Advantages

- Fast file transfer
- Reliable communication

Disadvantages

- Not secure without encryption

---

5. Email

Definition

Email is a system for sending and receiving digital messages over a network.

Protocols Used

Protocol	Function
SMTP	Sending email
POP3	Receiving email
IMAP	Accessing emails on server

Working

1. Sender writes email
2. SMTP sends email to server
3. Receiver downloads email

Advantages

- Fast communication
- Low cost
- Global access

6. TELNET

Definition

TELNET is a protocol used for **remote login to another computer** over a network.

Features

- Works on TCP
- Port 23
- Command-line interface

Disadvantages

- Not secure (data is not encrypted)
- Replaced by SSH

Example

Accessing a remote server terminal.

---

7. Domain Name System (DNS)

Definition

DNS is a system that converts **domain names into IP addresses**.

Working

1. User enters domain (e.g., google.com)
2. DNS server finds IP address
3. IP address is returned
4. Browser connects to server

Example

google.com → 142.250.190.78

Advantages

- Easy to remember names
- Faster navigation
- Scalable system

---

8. SNMP (Simple Network Management Protocol)

Definition

SNMP is a protocol used for **monitoring and managing network devices** like routers, switches, and servers.

Components

1. SNMP Manager
2. SNMP Agent

3. MIB (Management Information Base)

Working

1. Manager requests data
2. Agent collects information
3. Response is sent back

Functions

- Network monitoring
- Performance management
- Fault detection

Advantages

- Easy network management
- Real-time monitoring

Summary Table

Service	Function
WWW	Web browsing
HTTP	Web data transfer
FTP	File transfer
Email	Message communication

TELNET	Remote login
DNS	Name to IP conversion
SNMP	Network monitoring

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